Are We Moving in the Right Direction?

A customer-base analysis of cohort-based LTV at Studio North

Evidence from January 2024–December 2025

Observation cutoff: January 1, 2026 (exclusive)

Kenan-Flagler Business School

MBA 743: Managing Customer Data

Mod II 2026

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The report first assesses observed cohort progress, then explains the forecasting model and its implications for strategy. The appendix provides specifications, validation and the analytical handoff.

EXECUTIVE SUMMARY | MANAGEMENT REPORT | ILLUSTRATIVE BUSINESS

Studio North: is growth becoming more valuable?

Customer acquisition, monthly renewal and contribution economics | 24 monthly acquisition cohorts, 2024-01 through 2025-12; data cutoff 2026-01-01

THE BUSINESS AND THE DECISION

Studio North sells a monthly creative-work membership to independent professionals. Templates support immediate tasks; guided learning builds skills. Members choose standard or premium plans, may buy add-ons, and decide monthly whether to renew. Search and partner workshops acquire customers seeking time savings or stronger skills.

Management needs to know whether successive acquisition cohorts create more durable contribution at a cost the company can recover quickly enough to fund growth.

Customer economics are strengthening: lower blended CAC, higher contribution and more cash recovered over six months. Retention also improves within each channel–need group after excluding both experimental arms. Short campaign tests do not establish the return from a larger budget.

ACQUISITION EFFICIENCY
$108 → $81
observed blended CAC per acquired customer

CUSTOMER VALUE
$107 → $130
six-month contribution per customer

CASH RECOVERY
-$1 → $49
six-month cash after blended CAC

Matched-age comparison: January–June 2024 versus January–June 2025; age 5, six observed receipts; acquired-customer weights. Values above are rounded to dollars; the evidence table reports cents. Differences use unrounded values.

WHAT THE EVIDENCE ESTABLISHES

Table 1. Matched-age evidence on acquisition efficiency, customer value and cash recovery.

Evidence	Earlier	Later	Interpretation
Sixth-month retained share	52.2%	60.9%	Observed at age index 5
Blended CAC	$107.67	$80.53	Tracked spending / all acquired people
Six-month contribution / customer	$106.52	$129.98	After recorded variable costs
Six-month cash after acquisition	-$1.16	$49.46	Contribution less blended CAC
Ex-test retained share	52.2%	61.6%	Both experimental arms excluded
Ex-test CAC / cash	$107.67 / -$1.16	$85.31 / $45.80	Same joining windows; both test arms excluded

The later window includes 98 experimental buyers and $650 of test spending; the earlier window has none. Sensitivity rows exclude both arms and their receipts. Cash improves by $46.95 excluding tests, versus $50.62 including them. Differences are computed from unrounded values.

Initial need is informative: build skill customers show $40 more six-receipt contribution and 23 percentage points higher retention than save time customers; variation within each group remains substantial.

RECOMMENDED MANAGEMENT RESPONSE

Action. Marketing and Analytics should verify that controls stayed protected before using later outcomes; otherwise run a longer test. Finance should monitor the ex-test $45.80 six-month cash benchmark, not use it as an automatic causal gate. Product should measure time to first useful outcome and test a specific onboarding change; the current event flags do not identify the cause of cohort improvement. Incremental contribution after media per assigned prospect: search / save time: $1.91 over 31 days (95% interval $0.03 to $3.78); partner / build skill: -$0.34 over 30 days (95% interval -$2.33 to $1.65). 1 of 2 intervals include zero; Search is encouraging; neither short window establishes lifetime returns.

CONDITIONS FOR REVISITING THE DECISION

Revisit the growth plan if later cohorts' six-month cash recovery falls toward the earlier window, if a valid longer campaign comparison fails to cover media cost, or if young-cohort outcomes fall materially below forecast. Historical blended CAC does not establish the return from another advertising dollar.

Studio North and the customer journey

The membership business, customer needs and possible sources of value and friction.

The membership experience behind the economics

Studio North serves independent professionals seeking faster creative work or stronger skills.

Templates support immediate work; guided lessons build capabilities over repeated use. A monthly membership is valuable when those benefits recur often enough to justify the next payment. Optional workshops and add-ons expand spending but do not by themselves establish durable membership value.

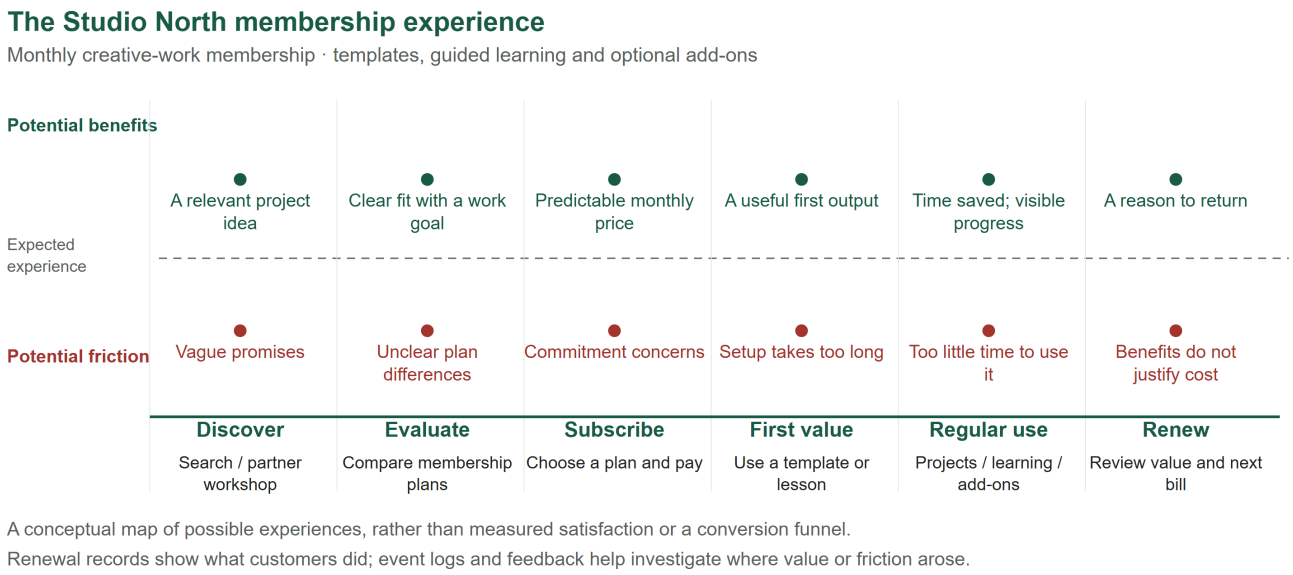


Figure 1. Conceptual. A broad conceptual journey through the Studio North membership. Benefits and friction points are possible experiences, not measured sentiment or population conversion rates.

POTENTIAL SOURCES OF RENEWAL FRICTION

An unclear plan choice can delay subscription. A difficult first task can prevent a member from experiencing value. Later, a lack of time or an unmet learning goal may weaken the reason to renew. These are business hypotheses for investigation; the source records do not establish their causes.

EVIDENCE AVAILABLE TO THE MANAGEMENT TEAM

The CRM records stated needs and baseline characteristics. Product events show activity, transactions show spending, renewal records show continuation, and feedback captures selected perceptions. Together, these sources support a more informative diagnosis than a renewal rate alone. Tables 20–27 classify all eight sources using the Universal Customer View. The appendix documents parameters, validation and reproducibility.

Observed cohort economics

Retention, contribution and acquisition-cost recovery at comparable customer ages.

Customer retention at comparable cohort ages

A joining cohort contains customers whose first successful payment occurred in the same month. Future ages remain unobserved.

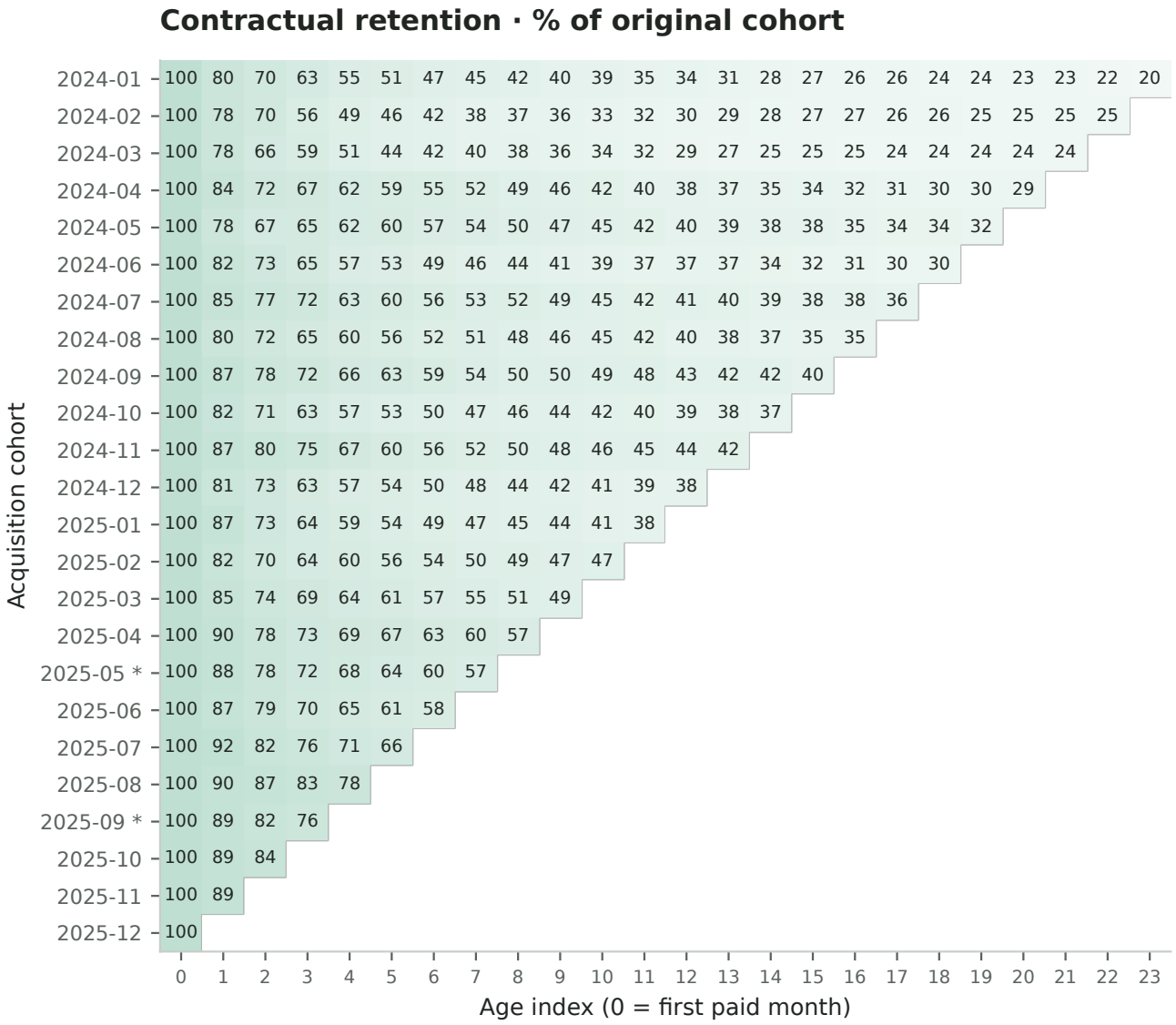


Figure 2. Observed. Customer retention for all 24 monthly acquisition cohorts and every observed age. Cells are percentages of the original cohort; blanks are unobserved. * marks test months, including buyers from both arms.

Saved SQL output: `cohort_totals.csv`. The full age range is 0–23; age 0 is the first paid month. Rows follow individual cohorts; columns compare successive cohorts at the same age (presentation background: Bouhnick). No missing future month is assigned zero retention.

COHORT AGE AND CALENDAR TIME

Every cohort starts at 100% because acquisition defines its original denominator. Newer cohorts have shorter histories. Their shorter lines and blank cells do not establish shorter customer lifetimes. Figure 4 shows the same full cohort sequence as curves; Figure 7 recombines cohorts in calendar time.

An improving business should acquire cohorts that retain more customers at comparable ages, rather than relying only on a larger inflow of new customers.

Contribution and acquisition-cost recovery

All 24 cohorts | Cash balance uses recorded contribution, tracked acquisition cost and the original acquired-customer denominator.

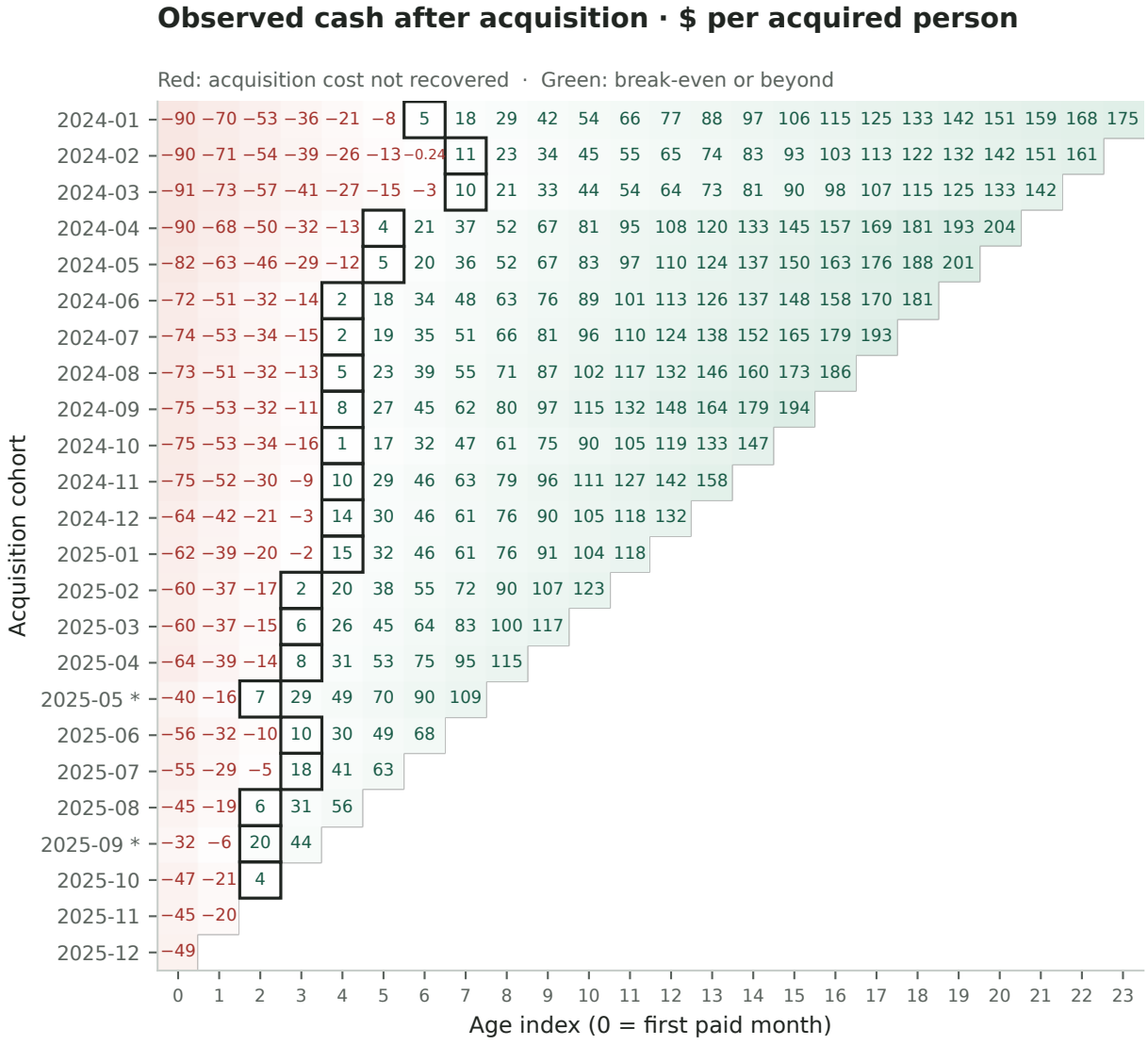


Figure 3. Observed. Cumulative cash after acquisition for every joining cohort, in dollars per original acquired customer. Red numerals indicate contribution below CAC; green numerals indicate break-even or beyond. An outline marks the first observed recovery of CAC. * marks test months, including both arms.

Saved SQL output: `cohort_cash_economics.csv`. Each cell equals cumulative net revenue less incurred variable costs and tracked acquisition spending, divided by original cohort size. The blank region is unobserved. This is undiscounted cash, not total company profit or discounted LTV.

ACQUISITION EFFICIENCY AND DURABILITY MUST MOVE TOGETHER

At age 5, six-month cash after acquisition changes from -$1.16 to $49.46 per original customer. The comparison uses January–June 2024 and the same joining months of 2025, weighted by acquired cohort size. Table 1 reports the matching retention, contribution and blended-CAC evidence. These are descriptive comparisons, not campaign effects.

> A new cohort may recover acquisition cost sooner because it costs less to acquire, generates more contribution per retained member, retains more members, or combines these changes. The retention and cash triangles together distinguish these sources of change.

Cohort trajectories

The relationship between member retention and revenue durability across the full sequence.

Retention and revenue across the cohort sequence

Every monthly acquisition cohort is shown. Shared axes and one chronological color scale keep the comparisons aligned.

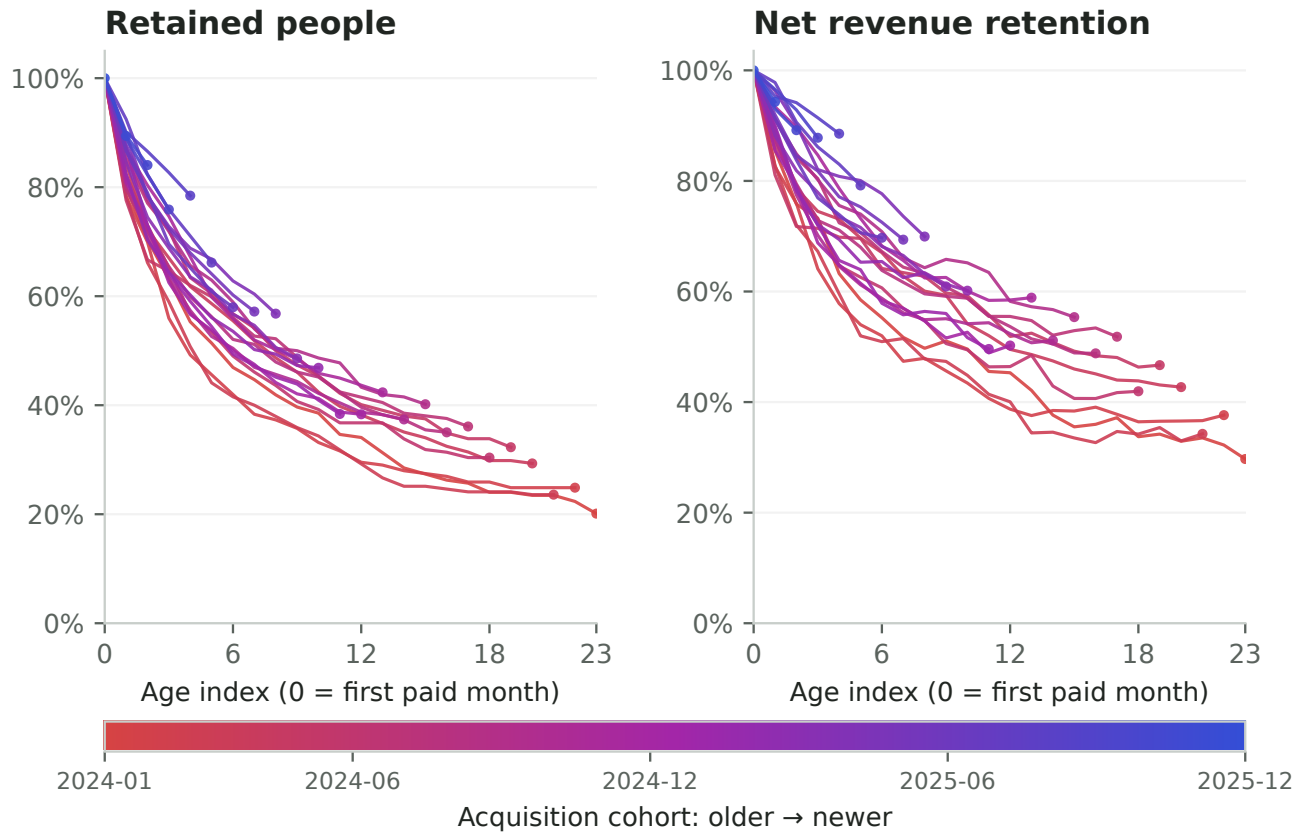


Figure 4. Observed. All 24 customer-retention and revenue-retention curves. Older cohorts are red and newer cohorts are blue; each line ends at its last observed month.

Retained share uses original cohort membership; revenue retention uses each cohort's first-period revenue. Revenue can change because of the number of renewing members, their plan mix or add-on spending. Product activity is not a substitute for a renewal record.

COHORT TRAJECTORIES AT COMPARABLE AGES

The vertical position of differently colored curves at the *same* age supports an aligned comparison. Later cohorts lying above earlier cohorts suggest improving retention; they need not dominate at every month. The chronological color key replaces an unwieldy 24-item legend. A curve with only its acquisition point has no observed renewal history.

CUSTOMER COUNTS AND REVENUE ANSWER DIFFERENT QUESTIONS

Higher revenue retention can coexist with customer losses when remaining members upgrade or buy add-ons. A retention improvement should therefore be checked against contribution and acquisition recovery in Figure 3, and against comparable customer groups in Figure 5.

The trajectory matters as well as the cutoff. A stronger first month does not establish longer customer life, while a customer-count decline can coexist with more spending by the members who remain.

Customer segmentation

Find useful cells with similar outcomes within each cell and different outcomes across cells, then connect them to decisions.

Similar within cells, different across cells

Compare candidate partitions on the same customers, outcome, horizon and number of cells.

THE OBJECTIVE IS A USEFUL PARTITION OF HETEROGENEITY

Choose a segmentation strategy that increases **within-cell homogeneity and across-cell heterogeneity** in a decision-relevant outcome. The MBA 743 grouping exercise holds willingness-to-pay values and cell counts fixed while changing who is grouped together (Tian, 2026, Topic 1, slides 37–39). Here the outcome is six-receipt contribution per originally acquired customer.

$$\underbrace{\mathrm{Var}(Y)}_{\text{fixed total variation}} = \underbrace{\sum_g w_g \,\mathrm{Var}(Y \mid g)}_{\text{within-cell variation}} + \underbrace{\sum_g w_g (\bar{Y}_g - \bar{Y})^2}_{\text{between-cell variation}}, \qquad w_g = \frac{n_g}{N}.$$

With the sample and outcome fixed, reducing weighted within-cell variance increases between-cell variance: they are the same objective. Population variances use the n_g divisor. Cell sizes supply the weights; a small cell does not receive the same weight as a large one.

Table 2. Four two-cell partitions of the same 4,278 mature, paid, non-experimental customers. Within-cell SD is the square root of customer-weighted within-cell variance. Between-cell share is between variance divided by total variance.

Two-cell strategy	Within-cell SD ($)	Between-cell share	After cohort centering
Stated need	65.60	8.31%	8.09%
Acquisition channel	68.24	0.78%	0.69%
Age band	68.50	0.03%	0.02%
Region	68.51	0.00%	0.00%

Saved output: `segmentation_objective.csv`, reconciled to the existing SQL profiles. All six receipts are observed; original customer denominators and zeros after exit are retained. Cohort centering subtracts each joining cohort's mean before repeating the variance comparison.

THE EVIDENCE FAVORS NEED, WITH SUBSTANTIAL RESIDUAL VARIATION

Stated need separates the group means most strongly among these four candidates: 8.31% of the six-receipt contribution variance lies between its cells, and 91.69% remains within them. After subtracting each joining cohort's mean, the between-cell share is 8.09%. This is a useful but modest descriptive improvement, not homogeneous customers or demonstrated intervention effects.

> Separation alone is insufficient: cells must also be recognizable, large enough to learn from and actionable. More cells can improve in-sample separation mechanically. Journey signals use different samples and timing, while channel-by-need cells support specific acquisition decisions; neither is automatically a better partition in this four-candidate comparison.

Customer differences and within-segment progress

Observed renewal trajectories by acquisition channel and stated need; each panel uses the same axes and all 24 cohorts.

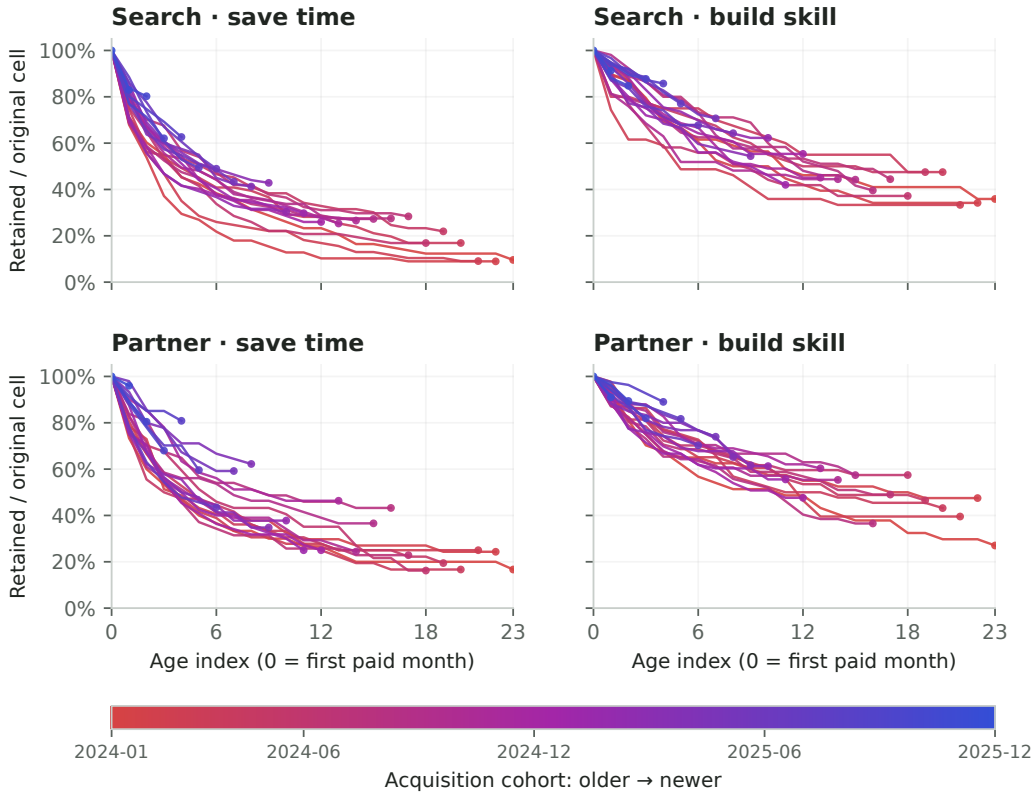


Figure 5. Descriptive comparison. The full cohort sequence within each channel-by-need group. The color key identifies acquisition month. Paid channels are shown; organic test-control cells are omitted.

Saved SQL output: `cohort_age_summary.csv`. Curves divide retained members by the original cohort–channel–need cell. No missing future observations are filled with zeros.

WITHIN-GROUP DIAGNOSIS

Excluding both experimental arms, age-5 paid retention is 52.2% for January–June 2024 versus 61.6% for the same 2025 joining months. Fixing the earlier channel-by-need mix gives 52.2% versus 60.2%; 4 of 4 paid groups improve. This removes changes in these groups' shares, not all customer differences or confounding. The curves above retain the full observed population; the table excludes both experimental arms.

Table 3. Sixth-month paid retention by original acquisition channel and baseline stated need; the same joining months one year apart.

Original channel / need	Jan–Jun 2024	Jan–Jun 2025	Change
Search / save time	39.0%	50.3%	+11.3 pp
Search / build skill	64.4%	69.9%	+5.5 pp
Partner / save time	45.3%	53.1%	+7.9 pp
Partner / build skill	69.7%	74.6%	+4.9 pp

Both experimental arms are excluded from this table and the fixed-mix calculation. The curves above include test buyers. Differences use unrounded values; the whole-company ex-test sensitivity appears in Table 1.

Segmentation accounts for part of the economic variation

Baseline characteristics define customer groups; their observed outcomes establish how useful those groups are for this decision.

ECONOMIC DIFFERENCES ACROSS CUSTOMER GROUPS

Customers whose initial stated need is build skill produce $39.59 more contribution over six receipts than those stating save time, on average. Their sixth-month retained share differs by 23.1 percentage points. This identifies an economically relevant customer distinction, not proof that changing a customer's need or acquisition channel would cause the same outcome.

Table 4. Six-receipt economics for the same mature paid customers, excluding both experimental arms. SD is the standard deviation of individual contribution within the stated-need group.

Initial stated need	Customers	Retained at age 5	Six-receipt contribution	Within-group SD
build skill	2,002	70.4%	$142.87	$66.79
save time	2,276	47.2%	$103.28	$64.54

Every comparison above uses the same mature paid customers, including zero contribution after churn. Differences use unrounded values. Both experimental arms are excluded. Table 28 compares four two-group segmentation strategies; cohort centering separates shared joining-month levels from within-group variation.

IMPLICATIONS FOR THE OPERATING PLAN

The more durable group supports a longer relationship and a different cash-recovery profile. The less durable group may still be attractive at an appropriate acquisition cost, with a service proposition suited to its shorter use cycle. Product and Marketing can distinguish immediate task completion from ongoing capability development when designing onboarding and renewal communications.

> Segmentation is a useful summary of market heterogeneity, not a claim that members of a group are identical. These comparisons identify where to investigate or test a decision; they do not estimate the effect of moving customers between groups.

Recorded early steps offer limited renewal separation

Post-acquisition diagnostics | Completion, core use and onboarding feedback observed in the first 27 days after payment.

RETURN TO THE CUSTOMER JOURNEY

The conceptual journey in Figure 1 distinguishes first value from regular use. Completion and core-use events show whether early steps occurred. Reports of difficulty getting started or finding features speak to first value; finding time speaks to regular use. None directly measures the effort of setup, the benefit of the first task, or the reason for cancellation.

Table 5. Descriptive outcomes by recorded early signal. Each row retains its original outcome-eligible customers, including subsequent nonrenewers. Contribution is per original customer at ages 1–5, excluding the acquisition month.

Recorded group	N, renewal 1	Retain 1	N, age 5	Retain 5	Ages 1–5 contribution
No core use recorded	792	85.5%	632	58.9%	$100.35
Core use recorded	4,618	85.4%	3,646	57.9%	$97.78
No completion recorded	690	86.1%	520	58.5%	$98.16
Completion recorded	4,720	85.4%	3,758	58.0%	$98.16
Finding features	179	83.8%	143	61.5%	$101.85
Finding time	162	84.0%	125	53.6%	$96.00
Getting started	162	85.2%	121	61.2%	$103.09
No recorded response	3,550	85.5%	2,852	57.7%	$97.21
No difficulty reported	1,357	85.8%	1,037	58.6%	$99.94

Saved SQL: `friction_segment_summary.csv`. Completion and core-use rows are separate, overlapping classifications; survey categories partition customers. First-renewal and age-5 columns have different maturity denominators. Refunds are assigned to their transaction month, as in the accounting tables.

WHAT SURVIVES AN ALIGNED COMPARISON

After standardizing acquisition year, channel and need, the first-renewal gap is +0.7 percentage points (95% interval -2.1 to +3.4) for no completion versus completion, and +0.1 percentage points (95% interval -2.4 to +2.7) for no core use versus core use. These estimates provide little separation, but their intervals do not establish equivalence. The signals record whether steps occurred, not the effort required or the value experienced.

SURVEY COVERAGE LIMITS THE DIAGNOSIS

Onboarding responses are recorded for 1,967 of 5,703 customers (34.5%). Invitations are unavailable, so no response combines uninvited people and people who did not answer. Difficulty categories are nominal and self-reported; a missing response is not a report of no difficulty.

These are recorded behaviours and selected perceptions, not identified causes of churn. The comparisons neither establish that friction is harmless nor estimate the effect of changing onboarding. Table 31 reports all adjusted first-renewal contrasts; Table 32 follows the five completion, core-use and reported-difficulty contrasts through their economic outcomes.

Renewal improves while recorded early steps vary little

All 24 joining cohorts, split by baseline stated need | Both randomized arms are excluded.

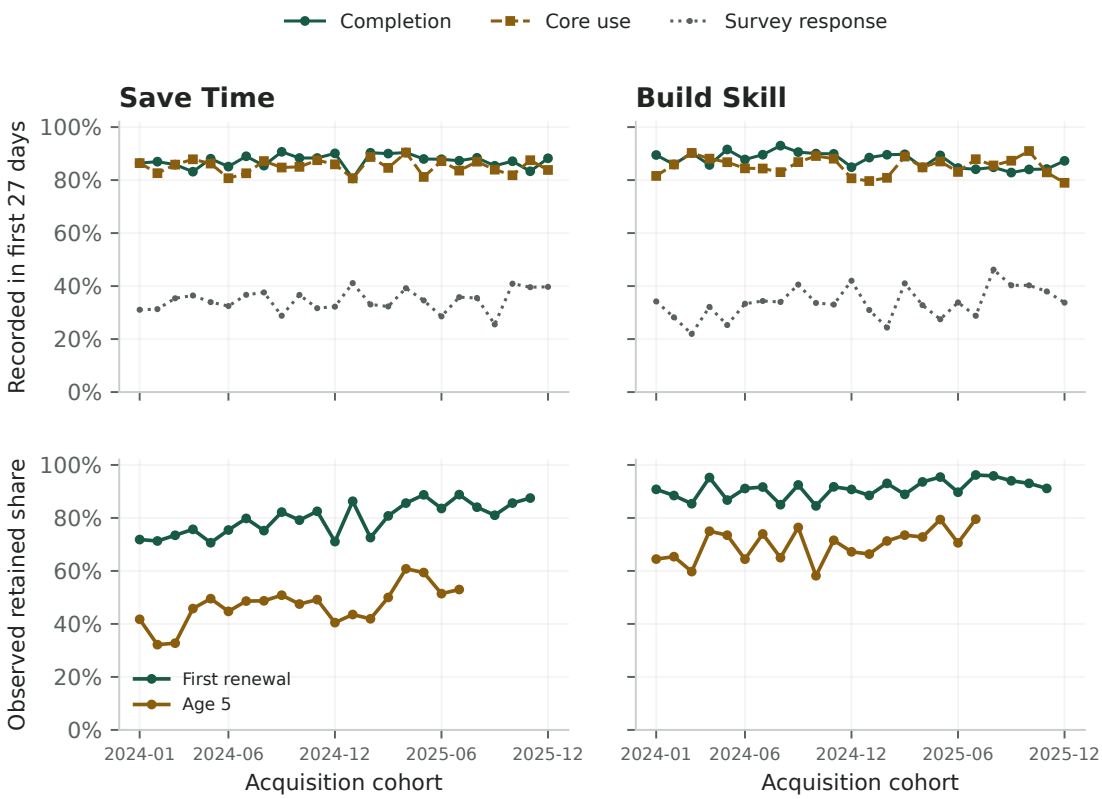


Figure 6. Descriptive comparison. Early completion, core use and survey response above; first-renewal and age-5 retention below. Outcomes stop at the last eligible cohort. All prevalence points have complete 27-day exposure.

Saved SQL: `friction_cohort_summary.csv`; detailed cohort–channel–need–signal counts: `friction_cohort_segments.csv`. Channels are pooled within each need in this figure. Test buyers are excluded throughout; remaining members in test months are retained. Unobserved outcomes are blank, not zero.

POPULATION AND TIMING

5,891 acquired customers minus 188 buyers assigned to either experimental arm leaves 5,703 paid non-experimental customers with a complete 27-day window. First renewal is observed for 5,410 (Jan 2024–Nov 2025); age 5 for 4,278 (Jan 2024–Jul 2025). All assigned prospects remain in the campaign analysis, including nonbuyers.

AN OPERATING QUESTION THE RECORDS CANNOT YET ANSWER

Relatively stable event prevalence alongside improving retention does not identify the mechanism of improvement. An easier experience could help members without changing whether a completion or core action is recorded. Conversely, a member may skip a recorded step and still obtain value. Product should measure time to first useful outcome and reasons for cancellation, then randomize a specific onboarding change within baseline need and acquisition channel.

Use initial need to distinguish the job members seek to accomplish. Use early activity and feedback to investigate that experience; the current signals do not justify automatic churn targeting or a claim that onboarding caused the cohort gains.

Company economics

How successive cohorts accumulate into an active membership base and operating contribution.

Retained cohorts determine how growth accumulates

The company is the sum of its acquisition cohorts at different ages, not the latest cohort scaled up.

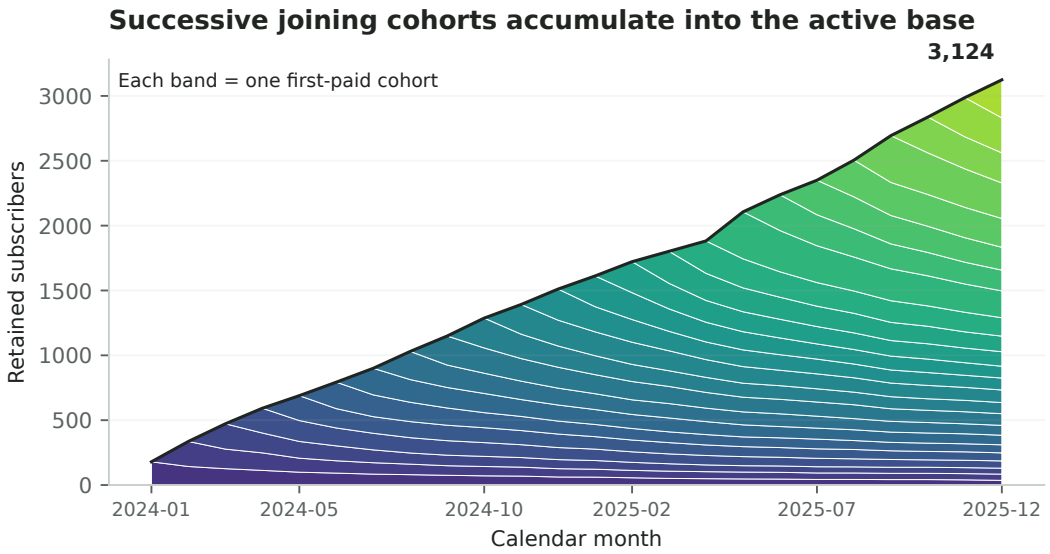


Figure 7. Observed. The company's retained membership base as the sum of all 24 acquisition cohorts in calendar time.

Saved SQL output: `company_month.csv`. Each layer is a joining cohort's active customers at the displayed calendar month. Layer totals are reconciled to independently saved company totals before plotting.

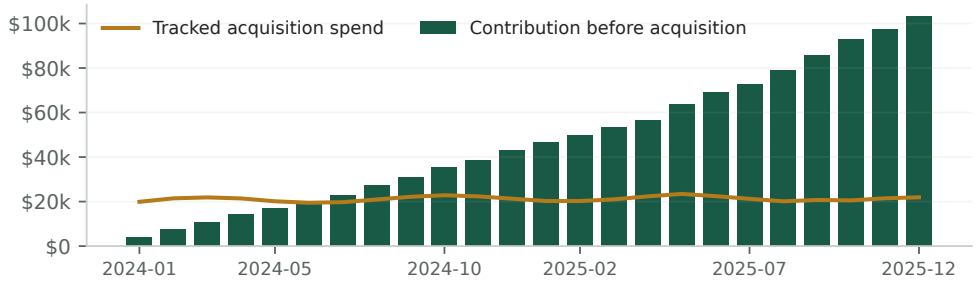


Figure 8. Observed. Monthly contribution before acquisition compared with tracked acquisition spending.

SCALE AND CASH RECOVERY ARE SEPARATE TESTS

The acquisition–retention–spending decomposition follows McCarthy, Fader and Hardie (2017). New acquisition adds customers immediately. Renewals preserve older layers and their contribution, reducing the replacement acquisition required to sustain scale. The lower chart compares recorded contribution with acquisition spending in each calendar month; it is not cumulative retained customers or a profit statement. In 2025-12, recorded contribution was $103,343 and acquisition spending was $21,915, leaving $81,428 before fixed costs, financing and taxes. This period result should be read alongside the age-aligned cohort comparison, rather than used as a substitute for it.

OPERATING IMPLICATION

The operating review benefits from both the common-age economics of each joining cohort and the company's accumulated contribution. A favorable acquisition metric can coexist with slower cash recovery when renewals weaken. A growing customer base can also conceal deterioration if management replaces departing customers with increasingly costly new ones.

> Cash payback is undiscounted: the first age at which cumulative contribution covers the acquisition investment. The LTV calculation that follows discounts expected monthly contribution and uses a common, explicit horizon.

Model-based forecasting of cohort trajectories

The renewal and revenue mechanisms that extend incomplete cohort histories to a common decision horizon.

How membership histories become cohort forecasts

Hierarchical shifted-Beta-Geometric retention + Gamma gross revenue; NumPyro NUTS

THE RENEWAL MECHANISM

The shifted-Beta-Geometric mechanism follows Fader and Hardie (2007). Each member has a persistent propensity to leave at a monthly renewal. Within a joining-cohort, channel and stated-need cell g, the model represents that heterogeneity with a Beta distribution. Members with higher churn propensity tend to leave earlier, so the surviving group's average churn falls even when no individual's underlying propensity changes.

$$p_{i,g} \sim \mathrm{Beta}(\alpha_g, \beta_g), \qquad h_g(t) = \frac{\alpha_g}{\alpha_g + \beta_g + t - 1}, \qquad t \geq 1.$$

Here $p_{i,g}$ is member i's latent monthly churn propensity and $h_g(t)$ is churn among members who reach renewal t. The first paid receipt is age zero; it establishes the acquired population and supplies no renewal trial. The count of customers who renew is modeled from the previous month's survivors:

$$N_{g,t} \mid N_{g,t-1} \sim \mathrm{Binomial}(N_{g,t-1}, 1 - h_g(t)) \,.$$

WHAT DIFFERS BETWEEN CUSTOMER GROUPS

The mean of the Beta distribution varies with baseline need, acquisition channel and joining-month trend, plus a partially pooled cohort effect. Its concentration varies with need and channel. Pooling lets a young cohort borrow evidence from older cohorts while allowing observed cohort differences. The model uses these baseline attributes; later activity and feedback are diagnostic evidence, rather than forecasting inputs.

REVENUE AND CONTRIBUTION

A Gamma model adapted from Orduz (2024) represents payment revenue. Its expected total is the number of retained members multiplied by expected spending per retained member:

$$\mathrm{E}[R_{g,a} \mid N_{g,a}] = N_{g,a}\, r_{g,a}.$$

Expected spending $r_{g,a}$ varies with customer age, joining month, need and channel. The payment target is after discounts and before refunds. Observed refund and variable-cost ratios translate future payment revenue into contribution. Actual contribution is kept wherever it is observed; only the unobserved remainder is forecast.

ESTIMATION AND SCOPE

NumPyro NUTS estimates the model from completed renewal and payment histories on CPU. Pending decisions and future ages remain unobserved. Cancellation is absorbing in this business. Forecasts start from the last actual survivor count and cover each cohort's first twelve monthly receipts. The appendix reports priors, parameter estimates and chronological validation; the separate Orduz-inspired neural model supplies a comparison, not executive LTV.

> The bands describe posterior parameter uncertainty conditional on the fitted model and fixed economic ratios. They exclude future customer-level variation, uncertain costs and business-regime changes. With no calendar-time term, a common disruption can affect multiple cohorts outside the displayed bands.

Extrapolation makes young cohorts comparable

A common twelve-receipt horizon combines observed contribution with an explicit forecast of the unobserved remainder.

WHAT THE MODEL CONTRIBUTES

Figure 9 applies the preceding model to cohorts with different amounts of observed renewal history.

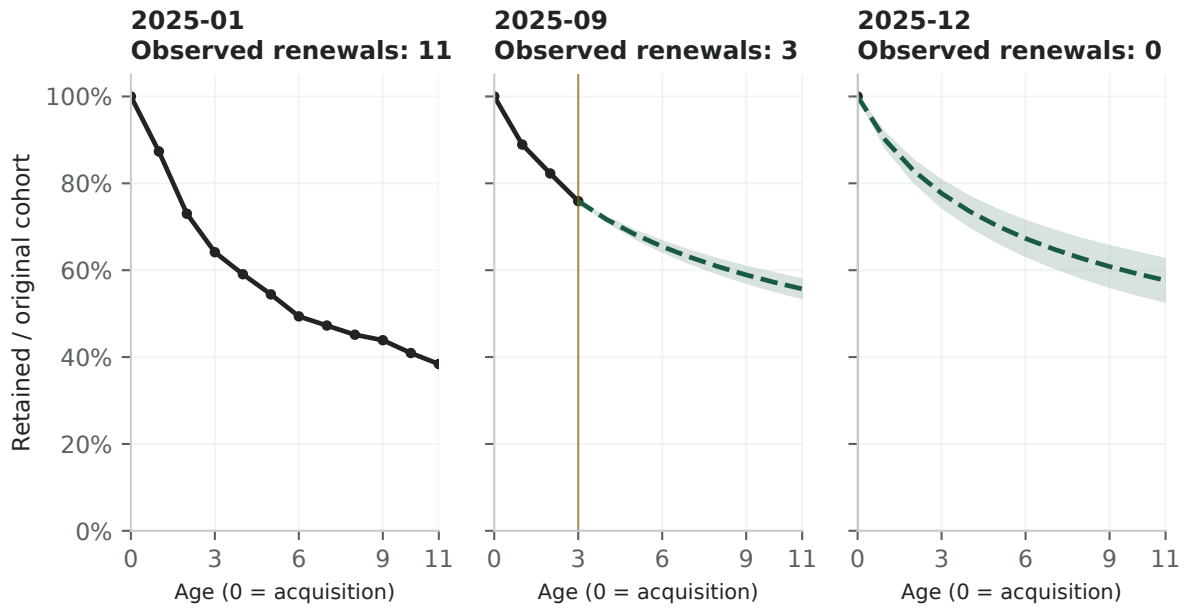


Figure 9. Observed and forecast retention for cohorts with complete, intermediate and minimal renewal histories. Solid points are observed; dashed continuations are forecasts. Examples are selected by history length, not outcome.

Saved model output: `forecast_curve.csv`. Original cohort denominators are preserved. Shading shows the 5th–95th posterior parameter range; the vertical marker identifies the observation boundary.

THE INFORMATION BEHIND EACH FORECAST

The mature cohort supplies a fully observed reference. The intermediate cohort combines its own renewal history with evidence from other cohorts. The newest cohort has only its acquisition month and therefore **zero observed renewal decisions**; its forecast depends primarily on comparable customers and model assumptions.

> Extrapolation supports an earlier assessment of cohort value; it does not turn future outcomes into observed facts. These bands describe parameter uncertainty in expected paths, excluding customer-level process variation and changes in the business. The preceding page explains the mechanism. Appendix A1–A2 gives the specification and estimates, A4–A6 reports model checks, and Section 7 defines valuation.

Customer value forecasts

Contribution valuation, acquisition-segment strategy and the assumptions behind immature-cohort LTV.

A common horizon connects renewal to acquisition economics

Twelve monthly receipts | Observed contribution is retained; only the unobserved remainder is forecast.

FROM THE FORECAST PATH TO VALUE

The model explained in Section 6 supplies the expected contribution path, preserving the retention heterogeneity emphasized by Fader and Hardie (2010). Discount each receipt, sum twelve receipts, then subtract the acquisition cost per original customer. The first paid receipt is age zero and is discounted by one month. Cash payback instead uses the first undiscounted crossing of cumulative contribution over CAC.

$$\text{net LTV}_H = \sum_{a=0}^{H-1} \frac{\text{E}[\text{contribution}_a \mid \text{observed history}]}{(1+d)^{a+1}} - \text{CAC}\,.$$

A separate revenue model estimates spending per retained member. Observed revenue and contribution are preserved; only the remaining receipts are forecast, with recorded cost and refund relationships held fixed. Values cover each cohort's first twelve monthly receipts, discounted at 1% per month. The model has no calendar-time effect: common disruptions can move cohorts together and are outside these parameter bands. The horizon covers twelve receipts, not an infinite lifetime. Bands describe uncertainty in expected value conditional on the model.

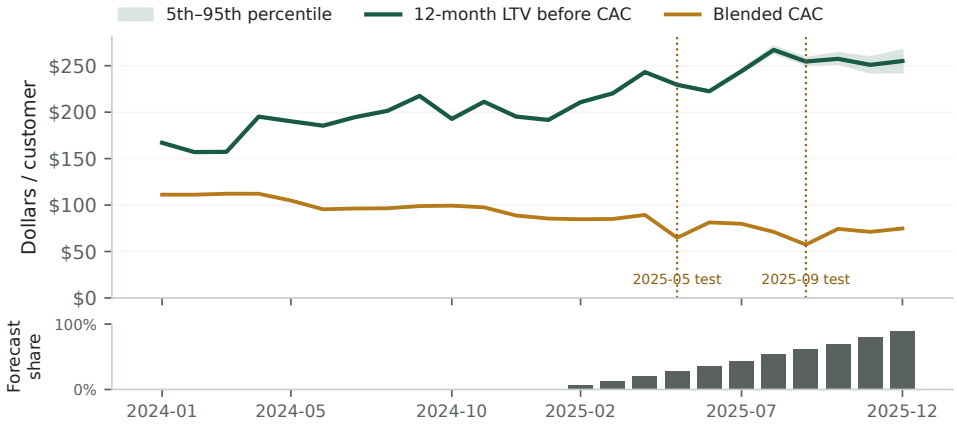


Figure 10. Observed and forecast. Common-horizon contribution LTV and blended CAC for all cohorts. The lower strip identifies the share of each LTV estimate that is forecast.

Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval. The lower strip shows the forecast share of common 12-month LTV. Dotted markers identify experiment months with organic control buyers in the denominator. Blended CAC divides tracked spending by all acquired customers; it is not the causal cost of the next customer.

MANAGEMENT INTERPRETATION

Among cohorts with all twelve receipts observed, net contribution value increases from $55.83 for 2024-01 to $106.18 for 2025-01. The newest cohort, 2025-12, is separate: $180 net value, 90% forecast, with a 5th–95th parameter range of $168 to $192. Its projection depends on the cohort-trend term and customer mix, with no observed renewal trial. In the historical one-receipt comparison, the model over-forecast value by $20.15; that miss illustrates risk beyond the parameter band.

Future net revenue and contribution shares of payment revenue use observed channel-by-need-by-age shares, with the same group's observed renewal ages (age>=1) as fallback. Shares are fixed assumptions, not causally identified or assigned posterior uncertainty. Predicted customer value is not experimentally identified campaign return. A favorable average LTV-to-CAC comparison still requires a cash-recovery constraint and evidence that the next acquisition investment creates additional value.

Acquisition groups differ in value and cash recovery

Post-estimation breakout | 2025-07, the latest cohort with six observed receipts and no experimental buyers.

VALUE AFTER A MEANINGFUL OBSERVED HISTORY

Table 6. Twelve-receipt contribution value per acquired customer. Value before CAC is discounted contribution; net value subtracts CAC once. Bands reflect 5th–95th parameter uncertainty in the unobserved expected remainder; realized receipts and CAC are fixed. They exclude customer-level variation and are not predictive ranges for realized cell outcomes.

Acquisition group	N	CAC	Value before CAC	Net value (5th–95th)	Forecast
Partner / build skill	71	$81	$299	$218 ($216–$220)	47%
Partner / save time	47	$72	$215	$143 ($141–$146)	41%
Search / build skill	61	$85	$284	$198 ($196–$201)	46%
Search / save time	87	$79	$187	$108 ($106–$110)	38%

Saved outputs: `strategic_segment_breakout.csv` and `postestimation_manifest.json`. Forecast share is the unobserved share of discounted contribution before CAC. Acquisition costs are supported by channel–need targeted spending. No costs or LTV are assigned to post-acquisition diagnostic groups.

SAME JOINING MONTH, ONE YEAR APART

Table 7. Each entry shows 2024-07 / 2025-07; N is the original acquired count. Retention is observed at age 5. Cash payback is observed for both cohorts and counts receipts from one. The newer cohort's twelve-receipt net value is partly forecast.

Acquisition group	N	Age-5 retained	Net 12-receipt value	Payback receipt
Partner / build skill	51 / 71	68.6% / 81.7%	$143 / $218	5 / 3
Partner / save time	35 / 47	37.1% / 59.6%	$40 / $143	7 / 4
Search / build skill	45 / 61	80.0% / 77.0%	$139 / $198	5 / 4
Search / save time	74 / 87	54.1% / 49.4%	$71 / $108	6 / 4

The earlier cohort has all twelve receipts observed. Differences between the two joining cohorts are descriptive; customer mix, the acquisition environment and other conditions may have changed. Selection uses maturity and experimental assignment, not unusually favorable outcomes.

Search's single-month age-5 shares changed by -3.0 percentage points for build skill; -4.6 percentage points for save time. These small cells do not establish a retention decline. The pooled January–June comparison remains the within-group retention evidence. Value can rise through lower CAC or higher spending per member without stronger retention.

DOES THE NEED RANKING RECUR ACROSS COHORTS?

Table 8. Build-skill minus save-time net value within the same channel and joining month. Pairs have at least six observed receipts and exclude any pair containing experimental buyers.

Channel	Positive gap / pairs	Median gap (range)	Fully observed: positive / pairs
Partner	19 / 19	$79 ($19 to $105)	13 / 13
Search	18 / 18	$76 ($29 to $121)	13 / 13

Saved output: `strategic_rank_stability.csv`. Some values include forecast remainders; the final column restricts to fully observed twelve-receipt pairs. This is descriptive consistency across historical cohorts, not independent forecast validation.

DECISION IMPLICATION

Within 2025-07, build-skill customers have mean net twelve-receipt value of $209, versus $120 for save-time customers, weighted by acquired cell size. This is a candidate distinction for acquisition and service tests; it does not show what the next advertising dollar will earn.

> The newest cohort remains a different decision problem: it has no observed renewal and is predominantly forecast. Use the mature-history breakout to frame a test, and the forecast-dependence display in Figure 10 to keep extrapolation risk visible. Neither average value nor average CAC establishes an optimal budget allocation.

The segment opportunity has a cash timeline

2025-07 | Solid points are observed; dashed continuations are forecasts from the same fitted model.

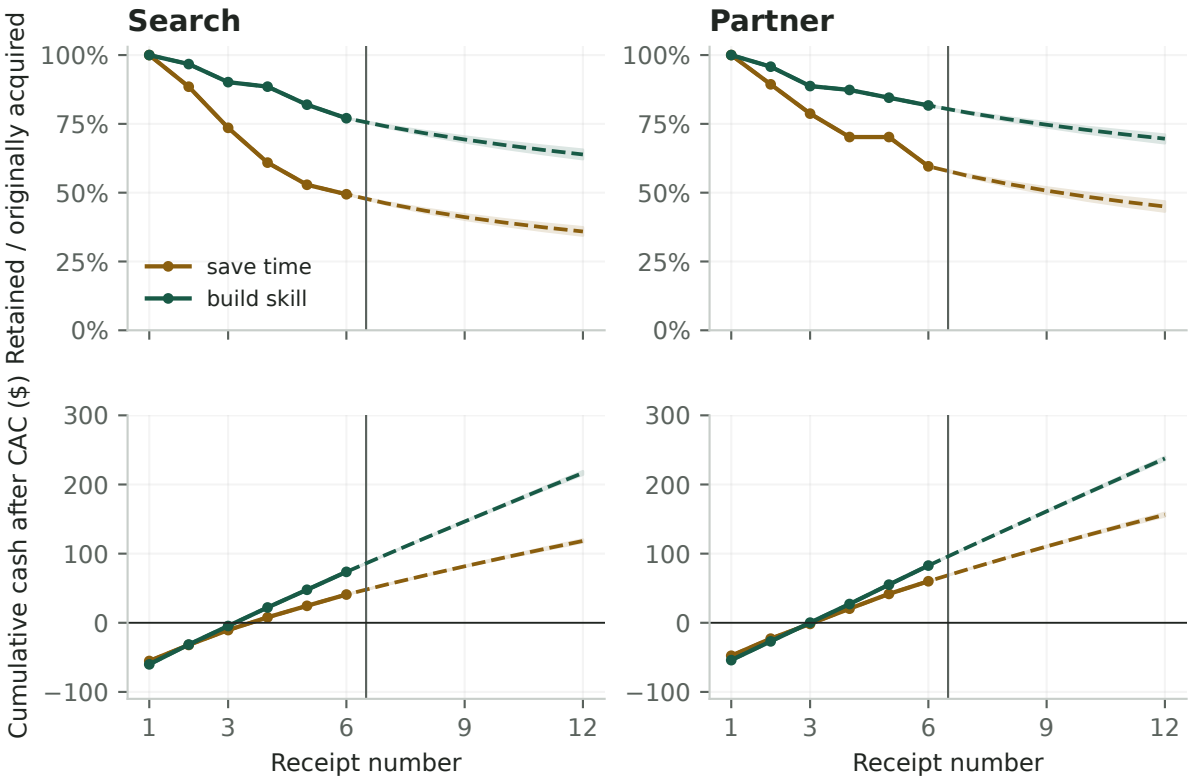


Figure 11. Observed and forecast acquisition-segment paths. The upper row shows retention; the lower row shows undiscounted cumulative contribution less CAC per original customer. The vertical marker separates six observed receipts from the remaining forecast.

Saved output: `segment_forecast_curves.csv`. Shading is the 5th–95th posterior parameter range. Original denominators remain fixed. Narrow conditional bands exclude future customer-level variation, uncertain costs and changes in the business. Latest non-experimental four-cell cohort with six observed receipts: 2025-07. Same-month prior-year comparator: 2024-07. All 98 acquisition-cell values and all 24 cohort paths reconcile to the saved baseline before selection; no refit.

TWO DISTINCT MANAGEMENT RESPONSES

Build-skill acquisition. Its more durable economics make this a candidate for a controlled acquisition test with a cash-payback constraint. The favorable partner cell does not establish that increasing partner spend will produce the same customers at the same CAC.

Save-time experience. Investigate whether a shorter use cycle reflects successful task completion, limited recurring need or an experience problem. The current event and survey flags do not distinguish those explanations. Test the proposition or onboarding change, then evaluate incremental contribution and cash over a stated follow-up window.

A segment can have high predicted buyer value while a campaign has little incremental effect. The randomized evidence in the next section determines what can be said about the tested campaigns; the breakout identifies where a new test may be informative.

Acquisition cost and incrementality

Reconcile campaign credit with the no-campaign baseline, then match acquisition cost to the value relevant to the decision.

One acquisition ledger, different decision quantities

Observed cohort economics and the causal return to an acquisition intervention answer different questions.

OBSERVED BLENDED CAC KEEPS THE COHORT ACCOUNTS CONSISTENT

$$\text{Observed blended CAC} = \frac{\text{recorded media + other acquisition spending}}{\text{all customers acquired in the matching cohort}}.$$

This report's earlier cohort exhibits use this accounting quantity with the contribution of those same customers. The denominator includes buyers who might have joined without the tested campaign. It is not incrementality-based. No global holdout identifies incremental CAC for all 24 cohorts. Charging every buyer spend per incremental buyer would deduct more than the recorded spending.

CAMPAIGN CREDIT IS A STARTING POINT FOR RECONCILIATION

For each holdout test, a simple attribution-style benchmark credits every treated buyer to the tested campaign: S_T/B_T, where S_T is treatment media spending and B_T is treated buyers. This is a transparent treatment-arm benchmark, not a newly estimated last-click metric. It overstates the number of customers caused by the campaign when some would buy in its absence.

INCREMENTAL CAC REMOVES THE COUNTERFACTUAL BASELINE

With $k = N_T/N_C$ to align assignment sizes,

$$\Delta B = B_T - kB_C, \qquad \Delta S = S_T - kS_C, \qquad \text{iCAC} = \frac{\Delta S}{\Delta B} \quad (\Delta B > 0).$$

Larsen (2019) motivates this distinction between attributed acquisitions, the background inflow and the return to additional spending. MBA 743 Topic 3, slides 6–10, develops the same cost-curve logic (Tian, 2026).

Control estimates acquisitions without the tested campaign in its eligible population and window. That background may reflect prior brand exposure and other marketing; it is not a company-wide count of purely organic customers. Each test compares zero media with $2.60 or $3.60 per assigned prospect. Its average incremental CAC describes that dose, population and window; it establishes neither a marginal cost curve nor a channel ranking.

Holdouts remove background acquisitions from campaign credit

The existing synthetic experiments already randomize acquisition exposure. Their assignments, purchases and spending are preserved.

FROM CREDITED BUYERS TO INCREMENTAL BUYERS

Each test assigns 250 prospects to campaign exposure and 250 to control. Search / save time is observed for May 1–June 1, 2025 (31 days); Partner / build skill for September 1–October 1 (30 days). End dates are exclusive. All assigned nonbuyers stay in the analysis.

Table 9. Control buyer counts estimate background acquisitions at the treatment-arm size. Both tests have $k = 1$; the implementation also supports unequal arms.

Test	Assigned T / C	T buyers	Expected baseline	Incremental buyers
Search / save time	250 / 250	73	25	48
Partner / build skill	250 / 250	61	29	32
Tested-program sum	500 / 500	134	54	80

Table 10. The same media ledger with two acquisition denominators. Credited CAC is treatment media divided by all treated buyers; background share is estimated no-campaign buyers divided by treated buyers.

Test	Extra media	Credited CAC	Background share	Media iCAC
Search / save time	$650.00	$8.90	34.2%	$13.54
Partner / build skill	$900.00	$14.75	47.5%	$28.13
Tested-program sum	$1,550.00	$11.57	40.3%	$19.38

Saved output: `cac_reconciliation.csv`; source: `experiment_summary.csv` and `experiment_effects.csv`. Control media and other acquisition-cost entries for these tested campaigns are zero. The displayed estimates are media iCAC; a fully loaded decision must also include any additional acquisition costs. Currency rounds half up to cents.

WHAT CHANGES IN THE RECONCILIATION

Search credits 73 buyers but induces an estimated 48; Partner credits 61 but induces 32. With zero control media spending, the correction multiplies credited CAC by $B_T/(B_T - B_C)$. If control spending is positive, both numerator and denominator must change; simply dividing attributed CAC by a lift fraction is insufficient.

> The tested-program row sums these two separately standardized effects and their costs. It describes this particular two-test program across different populations and windows. It cannot calibrate all acquisition spending, establish a universal channel multiplier or replace whole-company blended CAC.

A precise cost ledger can still produce an uncertain iCAC

Buyer lift is estimated. Dividing by a small or uncertain lift can make acquisition cost highly uncertain.

Table 11. Point estimates and approximate 95% intervals. Buyer-lift intervals use independent randomized arms; iCAC intervals invert the positive lift interval with media spending fixed.

Test	Incremental buyers (95%)	Media iCAC (95%)
Search / save time	48 (31.1 to 64.9)	$10.02 to $20.89
Partner / build skill	32 (15.4 to 48.6)	$18.52 to $58.45
Tested-program sum	80 (56.3 to 103.7)	$14.95 to $27.52

KEEP THE DENOMINATOR'S UNCERTAINTY VISIBLE

For buyer probabilities $\hat{p}_T = B_T/N_T$ and $\hat{p}_C = B_C/N_C$,

$$\mathrm{SE}(\Delta B) = N_T\sqrt{\frac{\hat{p}_T(1-\hat{p}_T)}{N_T} + \frac{\hat{p}_C(1-\hat{p}_C)}{N_C}}.$$

If the lift interval is $[L, U]$ with $L > 0$, the corresponding iCAC interval is $[\Delta S/U, \Delta S/L]$. If the lift interval crosses zero, there is no finite upper iCAC bound over its positive region. A zero or negative point estimate of lift does not supply a meaningful positive acquisition-cost estimate.

These are unadjusted normal approximations, conditional on fixed media spending and independent assignment units. The program row sums separately standardized effects and, assuming independent tests, their variances. Shared audiences or overlapping interventions would require a joint design and a different covariance calculation.

INCREMENTAL ACQUISITIONS DO NOT GUARANTEE A PROFITABLE CAMPAIGN

The all-assigned point decompositions are Search: $23.47 contribution per incremental acquisition versus $13.54 media iCAC, leaving $476.70 total net contribution; Partner: $25.47 contribution per incremental acquisition versus $28.13 media iCAC, leaving -$84.95 total net contribution. The value-to-cost ratios are 1.73 and 0.91. These contribution-per-incremental-buyer ratios are point decompositions; their intervals would require the joint value/lift distribution. The existing net-contribution intervals remain the inferential statements. A bounded iCAC does not establish profit: Partner's net-contribution interval includes zero. The following management exhibit retains both net-effect intervals and their 30/31-day horizons.

The control arms identify only the tested contrast. Estimating company-wide incremental CAC requires holdouts or coordinated experiments covering the relevant portfolio, with cross-channel overlap and substitution handled explicitly. Estimating a marginal cost curve additionally requires evidence across spending levels.

Match acquisition cost to the value being measured

Use the same population, intervention, cost scope, time origin and horizon on both sides of the decision.

FOR THE ACQUIRED COHORT: OBSERVED COST AND COHORT VALUE

The earlier exhibits combine average discounted contribution before CAC with observed blended CAC for the same acquired customers. Net cohort LTV subtracts that acquisition cost once. An LTV/CAC ratio uses value *before* CAC in the numerator. Do not subtract iCAC from the lab's net LTV, or divide that net LTV by iCAC and interpret it as a causal acquisition return (Tian, 2026, Topic 4, slides 3 and 31).

FOR AN ACQUISITION INTERVENTION: INCREMENTAL VALUE AND COST

Let V_H be discounted contribution before acquisition cost for every assigned prospect over a prespecified horizon H from assignment, including zeros. The campaign decision is

$$\text{Incremental net value}_H = N_T\left(\bar{V}_{T,H} - \bar{V}_{C,H}\right) - \Delta S_H.$$

Cost must include all additional acquisition spending relevant to this contrast. Dividing incremental contribution and cost by the same positive ΔB gives a consistent per-incremental-acquisition comparison: $\Delta V_H - \Delta S_H = \Delta B(\Delta V_H/\Delta B - \mathrm{iCAC})$. Count buyers and value to the same H under the same regime. However, that contribution ratio is the value of the intervention per additional acquisition; calling it the LTV of induced buyers requires assumptions, including monotone acquisition response and no change in spending or retention among people who would buy anyway.

WHAT THE CURRENT DATA CAN ESTABLISH

The existing experiments establish acquisition lift and short-run contribution effects. Controls are documented only through their prespecified 30/31-day windows. Later transactions do not establish continued control protection, so they do not identify a lifetime campaign effect. The cohort model forecasts twelve receipts from acquisition, a different clock from H after assignment. Multiplying conversion lift by average treated-buyer LTV would assume the unobserved induced buyers have that value and ignore possible effects on existing would-be buyers. The short-run effect is not a lower bound: pull-forward can let controls catch up, while retention can add value. A longer protected holdout could estimate discounted contribution for everyone assigned; extrapolation requires additional assumptions.

Use iCAC for the tested acquisition decision with matching incremental value. Keep observed blended CAC for the matching cohort accounts. The lab's cash-payback charts use *undiscounted* contribution; MBA 743 Topic 4, slides 30 and 35, teaches discounted contribution payback. These are distinct conventions, and the existing cash figures retain their stated convention.

Campaign incrementality and investment decisions

Randomized assignment includes all eligible prospects, including those who never purchase. Each test keeps its own control and follow-up window.

Table 12. Assigned prospects, buyers, recorded contribution and media spending by randomized arm.

Test	Arm	Assigned	Buyers	Contribution	Media
search / save time	control	250	25	$690.75	$0.00
search / save time	treatment	250	73	$1,817.45	$650.00
partner / build skill	control	250	29	$844.10	$0.00
partner / build skill	treatment	250	61	$1,659.15	$900.00

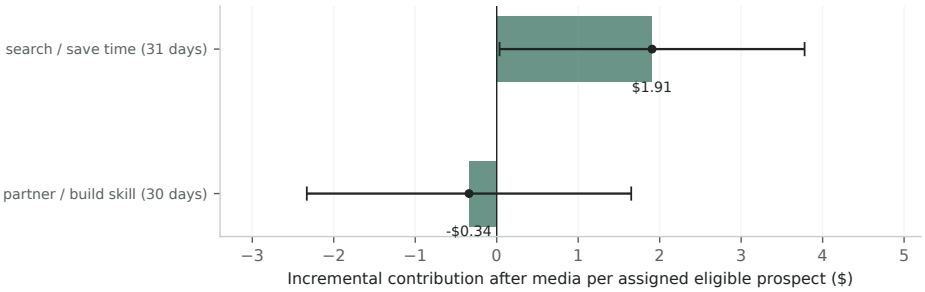


Figure 12. Randomized experiment. Incremental contribution after media per assigned prospect, with approximate 95% intervals over each test's recorded follow-up window.

Saved SQL outputs: `experiment_summary.csv` and `experiment_effects.csv`. Whiskers are approximate 95% intervals from all-assigned contribution variances, including nonbuyer zeros. Fixed media costs shift the mean, not its sampling variance. These 30/31-day windows do not measure lifetime campaign return. Later transactions exist, but the exports do not document whether controls remained protected after the prespecified follow-up.

THE SAME ECONOMICS APPLY TO TREATMENT AND CONTROL

For each arm, net revenue less recorded variable costs gives contribution. The difference in contribution per assigned eligible prospect is reduced by the treatment's additional media cost. Control purchases remain in the baseline rather than receiving campaign credit.

$$\Delta\text{contribution after media per prospect} = \frac{C_T - M_T}{N_T} - \frac{C_C - M_C}{N_C}.$$

MANAGEMENT ACTIONS AND ACCOUNTABILITY

Table 13. Accountable teams, next actions and the evidence required to revisit the recommendation.

Accountable team	Next action	Decision evidence
Marketing + Finance	Assess a larger budget using longer-window incremental contribution and cash recovery.	Age-aligned value and cash, alongside uncertainty in incremental returns.
Product + Operations	Instrument time to first useful outcome and test a specific onboarding change within baseline need groups.	Protected controls; subsequent renewal and contribution, including nonrenewers.
Marketing + Analytics	Verify control protection before extending the comparison; otherwise run a longer test.	Contribution after media per assigned prospect, including nonbuyers, with uncertainty.
Data + Analytics	Refresh source exports and review forecast errors before the next decision.	SQL reconciliation, model holdouts, parameter output and file hashes.

Experimental conversion lift multiplied by predicted buyer LTV is an assumption-based projection, not an established lifetime campaign effect. The tested population, outcome window and valuation assumptions must remain visible.

Technical and data appendix

Model parameters, historical validation, source definitions and the reproducible analytical handoff.

Parameter meanings and fitted estimates

Technical appendix | Parameter definitions, units, priors and fitted uncertainty.

LIKELIHOOD, HETEROGENEITY AND ECONOMIC CONVERSION

Each member has a persistent churn propensity. Within a cohort/need/channel cell, propensities follow one Beta distribution. After t-1 renewals, churn hazard is $\alpha/(\alpha+\beta+t-1)$; observed survivors follow a Binomial with the previous survivor count as its risk set. Mean propensity varies with need, channel, cohort trend and a partially pooled cohort effect. Concentration varies with need and channel. Variance is $\mu(1-\mu)/(\phi+1)$. Falling aggregate churn can reflect survivor selection, not improvement in each individual's loyalty. Gross payment revenue follows a Gamma distribution with shape proportional to observed retained members. Expected revenue per retained member varies with log age, cohort trend, need and channel. Observed age-aligned net and contribution shares translate future gross revenue into economics. Pending renewals and future ages are unobserved, not failures. Age zero establishes acquisition and supplies no renewal trial. Churn is absorbing in this business.

$$p_{i,g} \sim \mathrm{Beta}(\alpha_g, \beta_g), \qquad \alpha_g = \mu_g \phi_g, \qquad \beta_g = (1-\mu_g)\phi_g,$$
$$N_{g,t} \mid N_{g,t-1} \sim \mathrm{Binomial}(N_{g,t-1}, 1-h_g(t)), \quad h_g(t) = \frac{\alpha_g}{\alpha_g+\beta_g+t-1}.$$

g indexes a cohort–need–channel cell; $t = 1$ is its first renewal. Search and save-time customers are the reference groups. Transformations, estimation settings and uncertainty scope are documented on the next page.

Table 14. Structural parameter estimates, priors, posterior intervals and sampling diagnostics.

Parameter	Prior	Mean	5th–95th	R-hat	ESS
cohort_scale	HalfNormal(0.25)	0.080	0.017 to 0.144	1.002	1,023
concentration_beta[intercept]	Normal(log(4),1)	1.120	0.977 to 1.273	0.999	2,621
concentration_beta[build_skill]	Normal(0,0.5)	0.388	0.153 to 0.626	1.001	2,965
concentration_beta[partner]	Normal(0,0.5)	0.119	-0.105 to 0.351	1.000	2,575
concentration_beta[organic]	Normal(0,0.5)	0.050	-0.660 to 0.807	1.000	5,677
mean_beta[intercept]	Normal(-2.5,1)	-1.336	-1.419 to -1.254	1.001	2,414
mean_beta[build_skill]	Normal(0,0.5)	-0.886	-0.996 to -0.773	1.001	3,108
mean_beta[partner]	Normal(0,0.5)	-0.243	-0.353 to -0.133	1.001	2,540
mean_beta[organic]	Normal(0,0.5)	-0.259	-0.692 to 0.154	1.000	5,272
mean_beta[cohort_trend]	Normal(0,0.5)	-0.356	-0.443 to -0.274	1.001	3,175
revenue_beta[intercept]	Normal(-0.5,0.8)	-0.607	-0.612 to -0.602	0.999	7,826
revenue_beta[log1p_age]	Normal(0.1,0.8)	0.106	0.103 to 0.108	0.999	5,893
revenue_beta[cohort_trend]	Normal(0,0.8)	0.028	0.024 to 0.032	1.000	7,768
revenue_beta[build_skill]	Normal(0,0.8)	0.063	0.059 to 0.068	1.000	9,064
revenue_beta[partner]	Normal(0,0.8)	0.005	0.001 to 0.009	0.999	8,936
revenue_beta[organic]	Normal(0,0.8)	0.053	0.027 to 0.078	1.000	6,434
revenue_shape	LogNormal(log(4),0.8)	16.216	15.160 to 17.311	0.999	6,758

The 24 standardized cohort offsets z have posterior means from -1.204 to 1.004; multiply each draw by cohort_scale to obtain its log-odds shift. Their largest R-hat is 1.001; smallest effective sample size is 3,315. The complete parameter_summary.csv provides every cohort effect, prior, interval and interpretation.

At the coefficient means, Search / save time at centered cohort trend and zero offset has mean latent monthly churn mu=20.8% and concentration phi=3.06. The cohort-trend coefficient's 5th–95th interval is -0.443 to -0.274; it is below zero, supporting lower mean churn in later cohorts within this model. This is a model-based description, not a causal effect of acquisition date.

Posterior means and 5th–95th percentiles are read from the saved, hash-bound parameter output. Units and reference groups are stated with the parameter definitions; transformed parameters are not interchangeable with their latent log-scale values.

Estimation settings and the scope of uncertainty

Technical appendix | Parameter precision, predictive accuracy and causal identification are different properties.

ESTIMATION AND REPRODUCIBILITY

Table 15. Recorded inference settings and convergence diagnostics for the full-data fit.

Setting or diagnostic	Recorded value
Inference	NUTS
Chains / warmup / samples	4 / 1000 / 1000 per chain
Posterior forecast draws	400
Maximum R-hat / minimum ESS	1.002 / 1,023
Divergences	0
Seed / numerical backend	20261116 / cpu / float64

FEATURE TRANSFORMATIONS AND INTERPRETATION

Mean-churn coefficients use a logit link; concentration and revenue use log links. Search and save-time customers are the reference groups. Retention cohort trend is $(\text{index} - 11.5)/11.5$; revenue cohort trend is $(\text{index} - 11.5)/12$. Revenue log-age is $\log(1 + \text{age})$. Cohort index runs from 0 to 23. A coefficient therefore needs its link, scale and reference group before it has a managerial interpretation.

INTERVAL SCOPE AND REMAINING RISK

Posterior parameter uncertainty conditional on the single-Beta model and fixed future economics; not a full predictive interval.

One historical year-end origin, not proof of accuracy for every future cohort or business regime.

> Convergence diagnostics assess whether the fitted posterior was explored adequately. They do not establish that the model captures future customer behavior or identifies a campaign's causal effect. Historical holdouts test prediction using information available at the forecast origin.

METHOD SOURCES AND ADAPTATION

Retention extends Fader-Hardie sBG with mean/concentration parameterization, logit/log covariate links, a non-centered cohort effect on mean churn, grouped conditional Binomial renewal likelihood and NUTS estimation. Revenue adapts Orduz's Gamma structure but uses observed active counts in fitting, an additional per-member shape multiplier, a mean link without an age-by-cohort interaction, need/channel covariates and $100 scaling. Conditional on observed activity, the revenue and retention likelihoods have independent parameter blocks. The separate Orduz-inspired 4-hidden-unit Bayesian neural network is a retention comparison only; its cumulative-count likelihood is changed to conditional renewal trials.

The forecast component of customer contribution

Technical appendix | The same model forecasts retained customers, then converts that path into revenue and contribution.

FROM AN EXPECTED RENEWAL PATH TO CONTRIBUTION

Figure 9 introduces the extrapolation required for young cohorts. The revenue model applies expected spending per retained member to that renewal path. Recorded variable-cost and refund relationships translate revenue into contribution; acquisition cost is deducted once in net LTV.

OBSERVED EVIDENCE AND THE REMAINING FORECAST

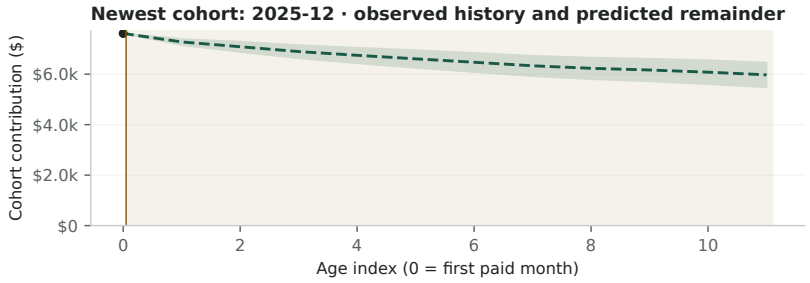


Figure 13. Observed and forecast. Observed contribution and the forecast remainder for the newest acquisition cohort.

Observed contribution is retained; only the unobserved remainder is predicted. The newest cohort's forecast relies on the cohort trend and customer mix until its own renewal history becomes available. Shading is the 5th–95th posterior parameter range. There is no calendar-time term; common shocks can affect multiple cohorts outside these bands.

These are expected customer paths conditional on fitted parameters. The bands exclude future customer-level Bernoulli variation, uncertain economic conversion and changes in the business regime. A narrow parameter band is not evidence that those risks are small.

Model fit and remaining timing patterns

Technical appendix | A model can describe observed counts closely and still miss the future.

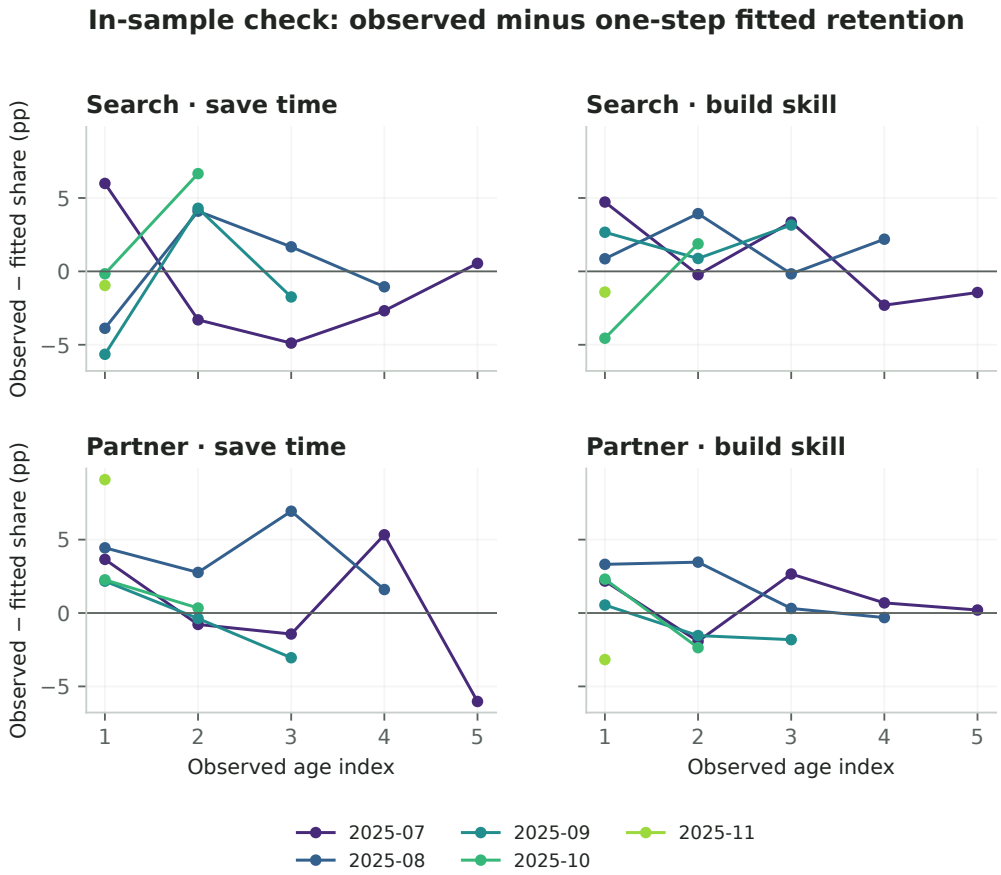


Figure 14. In-sample diagnostic. In-sample observed-minus-fitted retention residuals for the five newest cohorts with renewal history, by channel and need.

Saved SQL/model output: `model_fit_checks.csv`. These are in-sample, one-step fitted residuals conditional on previous observed customer counts. They are not held-out forecast errors.

WHERE CALENDAR-TIME FIT DEPARTS MOST

Table 16. The four largest absolute standardized calendar residuals among observed renewal risk sets.

Calendar month	Observed renewals	One-step fitted	Standardized residual
2025-04	1,632	1,665.4	-3.01
2025-03	1,555	1,584.4	-2.65
2024-10	1,057	1,034.5	+2.25
2025-08	2,222	2,199.1	+1.95

Four months with the largest absolute standardized calendar residual, selected from saved `calendar_fit_checks.csv`; only renewal ages 1 and older enter, not new acquisitions. These flag timing patterns, not causes, held-out errors or formal significance tests.

Historical prediction against withheld outcomes

Technical appendix | Forecasts are tested against outcomes withheld at historical forecast origins.

HELD-OUT PERFORMANCE

Training includes 2025-09; withheld months are 2025-10, 2025-11, 2025-12.

Withheld cells: 258. Retained-count mean absolute error: 1.22. Revenue weighted absolute percentage error: 5.9%. A separate training-only forecast checks complete 12-month values: LTV mean absolute error $4.02 versus $24.67 for its baseline. These tests evaluate prediction, not campaign effects.

Table 17. Prediction performance on historical holdouts, with explicit baseline comparisons and units.

Measure	Model	Baseline	Interpretation
Retained-count MAE	1.22	2.42	People per withheld cell
Retained-count WAPE	4.3%	Not reported	Retained customers, not revenue
Revenue WAPE	5.9%	12.8%	Withheld calendar months
12-month LTV MAE	$4.02	$24.67	Value per original customer

On the same 258 held-out cells, retained-count MAE is 1.215 for sBG, 1.335 for the Orduz-inspired 4-hidden-unit network, and 2.424 for the naive baseline. The network uses variational inference and a conditional monthly churn likelihood; it does not supply executive LTV. Revenue WAPE evaluates the main Gamma revenue model, not the retention-only network. The neural architecture is adapted from Orduz (2024); the bundled source-notebook edition and its provenance are listed in References.

WHAT THE COMPARISON CAN ESTABLISH

The model and each baseline use the same withheld outcome cells. Their errors are reported in the stated units and at the same forecast horizon. A better retained-count forecast need not produce the better revenue forecast, and predictive performance does not establish a campaign's causal effect.

Outcomes after the 2026-01-01 cutoff do not enter estimation, tuning or features. Forecasts for the next twelve calendar months can be evaluated as those results become available. That window differs from the valuation of each cohort's first twelve monthly receipts.

Observed and forecast components of historical value

Technical appendix | Training includes the calendar month beginning 2024-12-01; later calendar months are used only for scoring.

At the 2024-12 forecast origin, 11 cohorts had an unfinished 12-month value horizon. Payback had already occurred for 7; those cases are excluded from forecast-month MAE. The 4 remaining cases with both observed and predicted recovery have payback-month MAE 0.00. This is a small future-payback sample, not evidence of universally exact prediction.

Table 18. Cohort-level historical value and payback forecasts, separating already observed recovery from forecast recovery.

Cohort	Observed / forecast months	Actual LTV	Model LTV	Baseline LTV	Payback actual / model	Payback evaluation
2024-02	11 / 1	$157.01	$157.15	$157.26	8 / 8	Already observed
2024-03	10 / 2	$157.27	$157.30	$157.48	8 / 8	Already observed
2024-04	9 / 3	$195.15	$194.34	$192.87	6 / 6	Already observed
2024-05	8 / 4	$190.16	$189.17	$185.80	6 / 6	Already observed
2024-06	7 / 5	$185.43	$183.94	$176.44	5 / 5	Already observed
2024-07	6 / 6	$194.67	$197.94	$184.43	5 / 5	Already observed
2024-08	5 / 7	$201.40	$192.77	$176.54	5 / 5	Already observed
2024-09	4 / 8	$217.49	$212.32	$184.41	5 / 5	Future; scored
2024-10	3 / 9	$192.74	$194.95	$143.43	5 / 5	Future; scored
2024-11	2 / 10	$211.19	$212.53	$156.92	5 / 5	Future; scored
2024-12	1 / 11	$195.21	$215.37	$111.72	5 / 5	Future; scored

Saved model output: `valuation_backtest.csv`. LTV values are discounted contribution before blended CAC, per original acquired customer. “Observed / forecast” counts the receipts within the common 12-month horizon; payback month counts receipts from 1.

WHY TOTAL-VALUE AND FORECAST-TAIL ERRORS ARE THE SAME

Observed contribution is copied identically into the actual and predicted totals. The forecast covers only the remaining months. Consequently, total LTV error equals the error in the unobserved tail; the long observed prefix does not create additional predictive accuracy. Rows are ordered by the amount of history available at the origin.

THE BASELINE AND ITS WEAKEST-INFORMATION CASE

The naive baseline freezes the last three observed renewal rates and revenue per retained customer. A cohort with only its acquisition month has no observed renewal rate; the baseline uses an explicit 0.8 renewal fallback. Its value is an assumption, not an estimated renewal rate. The December row makes this low-information comparison visible. The baseline underpredicts all nine rows with at least three forecast receipts; its tendency to underpredict increases the apparent advantage of the fitted model.

WHAT THE PAYBACK FORECAST REPRESENTS

Forecast payback is the first crossing of acquisition cost by the **posterior-mean expected contribution path**. It is not the mean or median of a payback-time distribution, and it does not estimate the chance that a realized customer cohort recovers its cost by that month. Posterior parameter bands also exclude future customer-level process noise.

> The small set of future-payback cases is a narrow diagnostic. Zero month error should be assessed alongside contribution-value errors and incomplete histories; it does not establish universal forecast accuracy. Outcomes after the observation cutoff are excluded from these historical checks.

Source systems and analytical definitions

Technical appendix | Eight separate exports become tested analytical views in DuckDB; no database server is required.

Eight source exports; 24 monthly acquisition cohorts; cutoff 2026-01-01. Pending renewal outcomes are excluded from observed renewal trials. All randomized prospects remain in experimental denominators, including nonbuyers.

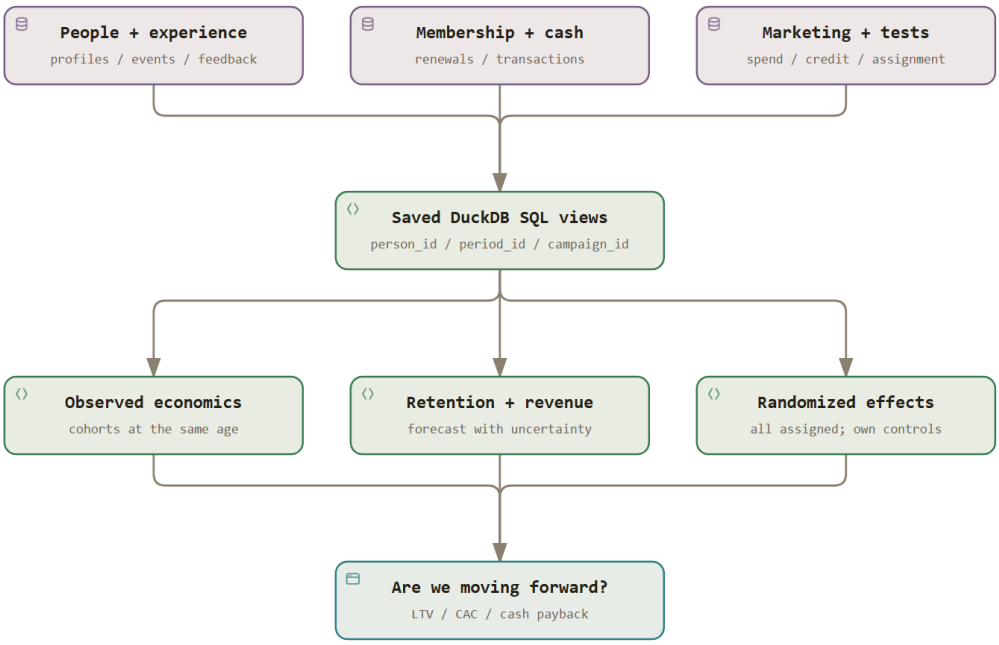


Figure 15. Eight source exports become analytical views and decision evidence through an auditable local pipeline.

The diagram is rendered by the bundled Archify package from original source specifications. The accompanying `diagrams/source-tables.html` and SVG show the full table-level schema and cardinality. Dataset classifications on the following pages list concrete fields.

KEYS DO NOT MAKE UNLIKE RECORDS INTERCHANGEABLE

A person can have many renewal periods, ledger transactions, product events and survey responses. Those tables must be aggregated at the intended grain before they are joined. Acquisition spending is a campaign-level record; it does not become a customer-level observation merely because it can be allocated across acquired customers. Experimental assignment defines an eligible population that includes noncustomers.

ANALYTICAL CONTRACTS

Table 19. Source-to-analysis definitions that preserve grain, timing, denominator and cost scope.

Definition	Contract
Joining cohort	Month of the first successful payment, derived from complete payment history.
Customer survival	Renewing contractual membership; original cohort size is the retained-share denominator.
Revenue and contribution	Ledger payments less discounts/refunds; contribution also subtracts recorded variable costs.
Acquisition investment	Tracked spend, including campaigns with no attributed acquisitions; no invented organic-source CAC.
Experimental unit	Every assigned eligible prospect, including nonbuyers and control buyers, within the prespecified follow-up window.
Unknown future	Right-censored ages stay unobserved. Missing survey responses remain missing perceptions.

Source hashes, saved SQL and downstream manifests bind the evidence chain. Numeric tables, editable LaTeX, source diagrams and the maintained style package accompany the report.

Source records, keys and cardinality

A3 landscape foldout | The eight raw exports remain separate. Campaign metadata is a derived lookup for a nonunique campaign-spend key.

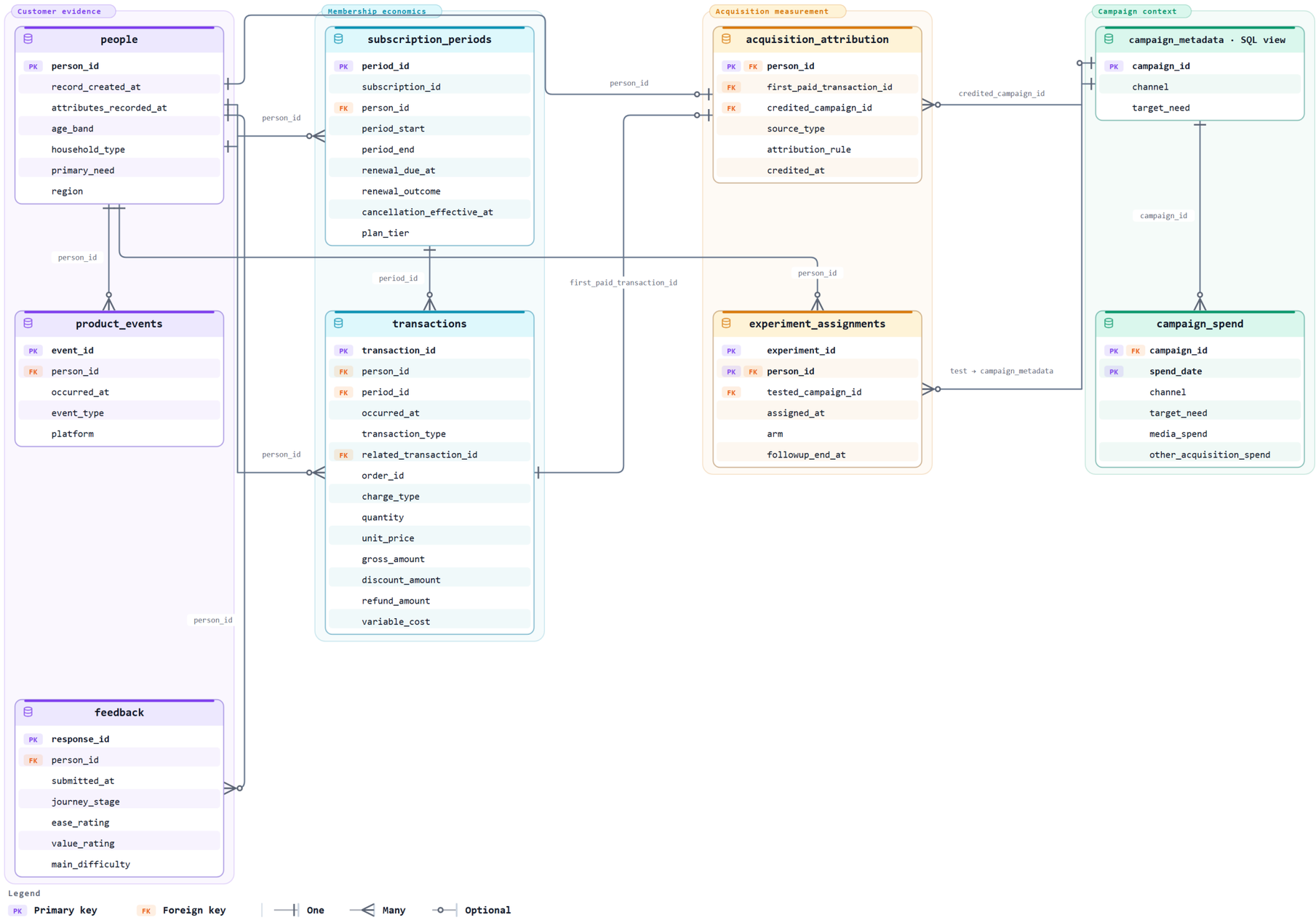


Figure 16. Source tables, join keys and cardinality, including the derived unique campaign lookup.

Native Archify schema; editable SVG/HTML: diagrams/source-tables. Refunds link transactions.related_transaction_id to transactions.transaction_id within the same ledger. Campaign foreign keys target unique campaign_metadata; campaign_spend uses campaign_id + spend_date. One-to-many sources are aggregated to the intended analytical grain before joining.

Source-level evidence in the Universal Customer View

Technical appendix | Context is marked explicitly; the categories describe evidence, not proof of an individual action or attitude.

Identifiers: Keys and profile descriptors identify or link people, groups and records; not every identifier is unique. Engagement: Observed customer interactions and their measurement context. Firm marketing activity and attribution metadata are retained as explicitly marked contextual evidence; they do not establish individual exposure, action or a causal effect. Relationship: Contractual and commercial arrangements, customer status, and explicitly marked operational context for relationship economics. Perception: Customer-reported views or intentions, distinguished from behavior and analyst inference.

Table 20. Universal Customer View: people.csv.

CRM | One person, including eligible prospects who never purchase.

Identifiers	Engagement	Relationship	Perception

`person_id`
`age_band`
`household_type`
`region`
	Not observed	`record_created_at`†
		`attributes_recorded_at`†
			`primary_need`

record_created_at: CRM relationship provenance; registration does not establish a paid customer. attributes_recorded_at: Profile provenance: verifies attributes were available before acquisition or assignment.

Table 21. Universal Customer View: subscription_periods.csv.

Renewal system | One scheduled monthly cycle; a paid outcome establishes retained membership for that cycle.

Identifiers	Engagement	Relationship	Perception

`period_id`
`subscription_id`
`person_id`
	Not observed	`period_start`
		`period_end`
		`renewal_due_at`
		`renewal_outcome`
		`cancellation_effective_at`
		`plan_tier`
			Not observed

Table 22. Universal Customer View: transactions.csv.

Finance | One payment or refund event; each payment is one order and refunds reuse its order ID.

Identifiers	Engagement	Relationship	Perception

`transaction_id`
`person_id`
`period_id`
`related_transaction_id`
`order_id`
	`occurred_at`
	`transaction_type`
	`charge_type`
	`quantity`
	`gross_amount`
	`discount_amount`
	`refund_amount`
		`unit_price`†
		`variable_cost`†
			Not observed

unit_price: The item's commercial price term is Relationship context; billed quantity and realized amounts record transaction Engagement. variable_cost: Firm-side accounting context for relationship economics; unlike billed amounts, it does not record a customer payment or reported perception.

Each raw field appears once in its dataset's classification. Source-specific rationale is preserved in `diagrams/ucv-tables.json`. A category marked not observed is a data gap, not an invitation to infer a missing attitude or relationship.

Customer activity, response and expressed perceptions

Technical appendix | Campaign spending and randomized assignment describe different objects from a customer's behavior or survey response.

Table 23. Universal Customer View: product_events.csv.

Product and support | One observed product/support interaction; no event does not mean no subscription.

Identifiers	Engagement	Relationship	Perception

`event_id`
`person_id`
	`occurred_at`
	`event_type`
	`platform`
		Not observed	Not observed

Table 24. Universal Customer View: feedback.csv.

Customer research | One completed linked survey response; nonresponse is missing, not a zero rating.

Identifiers	Engagement	Relationship	Perception

`response_id`
`person_id`
	`submitted_at`
	`journey_stage`†
		Not observed	`ease_rating`
			`value_rating`
			`main_difficulty`

journey_stage: Context of the response, not a measured universal customer path.

Table 25. Universal Customer View: campaign_spend.csv.

Marketing spending | One campaign per spending date; aggregate before joining to customer outcomes.

Identifiers	Engagement	Relationship	Perception
`campaign_id` `target_need`†	`spend_date`† `channel`† `media_spend`† `other_acquisition_spend`†	Not observed	Not observed

target_need: Identifies the campaign's target group; unlike a person's primary_need, it is not an individually stated intention. Channel describes the firm's marketing activity, so it is Engagement context instead. spend_date: Timing of firm marketing activity, not a customer touchpoint timestamp. channel: Context of firm marketing activity; it does not prove an individual's exposure. media_spend: Firm-side campaign spending is retained as Engagement context, not individual engagement or evidence of exposure. other_acquisition_spend: Firm-side creative/vendor costs are Engagement context, not observed customer activity; aggregate once before joining customers.

Marketing credit and experimental assignment

Technical appendix | A dagger marks contextual placement. It is not part of a source field's name.

Table 26. Universal Customer View: acquisition_attribution.csv.

Marketing attribution | One acquired person and the recorded credit for the first successful payment.

Identifiers	Engagement	Relationship	Perception
person_id first_paid_transaction_id credited_campaign_id	source_type† credited_at† attribution_rule†	Not observed	Not observed

source_type: Recorded acquisition-path credit; an attribution label does not establish incrementality. credited_at: When credit was recorded, not necessarily the acquisition or exposure time. attribution_rule: Measurement metadata for engagement credit, not a customer action itself.

Table 27. Universal Customer View: experiment_assignments.csv.

Experiment platform | One eligible person per randomized experiment, including all nonbuyers.

Identifiers	Engagement	Relationship	Perception
experiment_id person_id tested_campaign_id	Not observed	assigned_at† arm† followup_end_at†	Not observed

assigned_at: Experimental relationship/design metadata; assignment is not observed exposure or purchase. arm: Randomized intervention assignment; control nonbuyers must remain in the denominator. followup_end_at: Prespecified measurement window, not a subscription end date.

SOURCES AND MODEL ADAPTATIONS

Juan Orduz, Revenue Retention and Customer Lifetime Value with NumPyro; Peter Fader and Bruce Hardie, contractual-customer retention and shifted-Beta-Geometric models; Gil Bouhnick, Cohort Analysis. Exact links, adaptation boundaries and licensing are retained in the repository source documentation. Archify is pinned and redistributed with its license notices.

All graphics in this report are generated from this repository's original synthetic records or original diagram specifications. The archived Orduz source notebook is redistributed under its repository's MIT license with attribution and the license notice. Other third-party screenshots and private course materials are not included.

The explanatory value of baseline customer groups

Technical appendix | Identical customers and outcome windows make the four segmentation strategies comparable.

DISCRIMINATION ACROSS TWO-GROUP SEGMENTATION STRATEGIES

Among four two-group strategies on the same 4,278 paid customers, primary need has the lowest within-group variance in six-receipt contribution (4,303.5 dollars squared). It reduces unexplained variation by 8.3% relative to no segmentation; after centering outcomes within joining month, the reduction is 8.1%. Substantial individual variation remains within each group.

Table 28. Four strategies, each with two groups. Variance reductions use the corresponding unsegmented outcome variance as the denominator.

Two-group strategy	Within variance ($ squared)	Reduction	Cohort-centered reduction	Smallest group
acquisition channel	4,657.1	0.8%	0.7%	1,829
age band	4,692.2	0.0%	0.0%	2,110
primary need	4,303.5	8.3%	8.1%	2,002
region	4,693.6	0.0%	0.0%	2,069

Saved SQL outputs: `segmentation_profile.csv` and `segmentation_comparison.csv`. Every strategy uses the same mature paid customers and six-receipt horizon. Cohort centering subtracts each joining month's mean contribution before calculating within-group variance; it removes that month's shared outcome level without fitting a model.

WHAT THE VARIANCE COMPARISON ADDS

A segmentation can produce an actionable difference in average retention and dollars while leaving substantial variation among its members. The main report presents those economic differences first. This diagnostic asks how much individual contribution variation each two-group strategy accounts for; it does not establish a causal effect or predictive accuracy outside the observed sample.

> Cohort centering removes the mean level of each joining month. It does not adjust for every confounder. The channel-by-need four-cell comparison is presented separately and is not ranked against two-group strategies.

Timing, coverage and baseline differences

Diagnostic appendix | Exposure is [first payment, first payment + 27 days); the outcome follows that window.

HOW EACH JOURNEY HYPOTHESIS MEETS THE RECORDS

Table 29. A conceptual friction is not the same object as its available recorded proxy.

Journey stage	Available evidence	Interpretation limit
Discover / evaluate	Acquisition channel, chosen plan and randomized campaign assignment	Buyer mix is observed; confusion or commitment concerns are not.
First value	Completion and core-use events; onboarding feedback within 27 days	Completion timestamps share a fixed offset. They do not measure setup duration or a useful outcome.
Regular use	Finding-time answers at onboarding; separate ongoing-use feedback later	The later wave follows renewals and reflects a selected population. It is excluded from early groups.
Renew	Subscription renewal outcome	Continuation is observed. Unrecorded cancellation requests and reasons are not.

BASELINE COMPOSITION OF THE DIAGNOSTIC GROUPS

Table 30. Composition of diagnostic groups. Skill means build-skill need; premium is the initial plan. Discount and age-0 contribution are mean dollars; age-0 contribution overlaps exposure and is context, not a pre-exposure control.

Recorded group	N	Partner	Skill	2025	Premium	Discount	Age-0 contrib.
No completion recorded	726	47%	48%	59%	18%	$1.14	$23.82
Completion recorded	4,977	43%	48%	56%	20%	$1.14	$24.31
No core use recorded	847	43%	48%	57%	21%	$0.99	$24.46
Core use recorded	4,856	44%	48%	56%	19%	$1.16	$24.21
Finding features	182	46%	47%	59%	16%	$1.15	$24.17
Finding time	174	44%	51%	60%	25%	$0.89	$25.46
Getting started	174	39%	45%	59%	21%	$1.26	$24.25
No recorded response	3,736	44%	48%	56%	19%	$1.13	$24.16
No difficulty reported	1,437	43%	48%	57%	21%	$1.16	$24.34

Saved output: `friction_balance.csv`. All complete-exposure non-experimental paid customers. Differences in baseline composition motivate stratification but do not exhaust possible confounding.

REFUNDS AND THE TIMING BOUNDARY

The exposure window contains 342 refund records, 0 effective cancellations and 0 period ends. Primary comparisons retain refunding customers. Excluding customers with a refund gives first-renewal adjusted gaps of +1.3 pp (-1.5 to +4.1) for no completion versus completion; +0.1 pp (-2.6 to +2.8) for no core use versus core use. Refunds do not establish exit. Cancellation-request times are unavailable, so the window precedes recorded renewal outcomes, not necessarily the customer's decision.

Saved timing and source audits: `friction_timing_audit.csv`, `friction_source_audit.csv`. Onboarding-start events precede payment and are not diagnostic features. The earliest later renewal is more than 27 days after payment. No age-1–5 transaction precedes the feature boundary.

Aligned contrasts retain substantial uncertainty

Descriptive standardization | Event contrasts use acquisition year; survey contrasts use acquisition month. Both retain channel and initial need.

FIRST RENEWAL: THE PRIMARY OUTCOME

Table 31. Target minus reference, in percentage points. Counts, pooled contrasts and adjusted contrasts use the same common-support strata. Coverage is the share of the full outcome-eligible population in those strata; weights reflect that population's composition within support.

Comparison	N: target / ref.	Crude pp	Adjusted pp	95% interval	Coverage
No completion vs. completion	690 / 4,720	+0.7	+0.7	-2.1 to +3.4	100%
No core use vs. core use	792 / 4,618	+0.0	+0.1	-2.4 to +2.7	100%
Support contact vs. none recorded	374 / 5,036	-0.7	-0.9	-4.6 to +2.9	100%
Getting started vs. no difficulty	139 / 713	-1.4	-1.1	-7.9 to +5.6	53%
Finding features vs. no difficulty	152 / 793	-3.6	-3.0	-9.9 to +4.0	62%
Finding time vs. no difficulty	139 / 760	-1.5	-2.6	-10.0 to +4.9	57%
No response vs. response	3,550 / 1,860	+0.1	+0.6	-1.4 to +2.6	100%

Saved output: `friction_contrasts.csv`. Each included stratum requires at least two members in both groups. Normal 95% intervals use within-group sample variances and independent people, treating weights as fixed. They are approximate, unadjusted for multiple comparisons and not posterior intervals. A small estimate or an interval spanning zero does not establish equivalence.

SIX-MONTH ECONOMIC FOLLOW-THROUGH

Table 32. Adjusted age-5 retention and ages 1–5 contribution contrasts, with approximate 95% intervals. Original mature-customer denominators retain zeros after churn. Common-support coverage can differ from the first-renewal comparison and is saved with every estimate.

Comparison	Age-5 gap, pp (95%)	Ages 1–5 contribution gap (95%)	Coverage
No completion vs. completion	+0.6 (-3.8 to +5.1)	$0.23 (-$5.49 to $5.96)	100%
No core use vs. core use	+1.1 (-2.9 to +5.1)	$2.77 (-$2.56 to $8.09)	100%
Getting started vs. no difficulty	-1.0 (-11.4 to +9.5)	$3.05 (-$9.76 to $15.85)	45%
Finding features vs. no difficulty	+2.4 (-7.8 to +12.7)	$0.58 (-$12.43 to $13.59)	55%
Finding time vs. no difficulty	-4.2 (-15.8 to +7.5)	-$7.04 (-$23.24 to $9.15)	57%

WHAT ADJUSTMENT DOES AND DOES NOT RESOLVE

Year–channel–need standardization for event groups leaves within-year cohort differences. Survey contrasts instead align joining month, channel and need, so months without a recorded response cannot enter the comparison. Sparse survey groups lose coverage; results apply to supported cells. Neither adjustment resolves plan selection, unmeasured experience or selective survey participation. No recorded response retains unknown invitation status.

> The figures and tables describe associations in the released observation window. They do not establish a friction effect, an individual targeting rule or prediction on future cohorts. A randomized intervention with a protected control is needed to estimate whether reducing a specific friction changes retention and contribution.

Knowledge assets and a reproducible handoff

Skills specify decisions and stage gates; reference assets explain methods; Python packages execute the transformations and models.

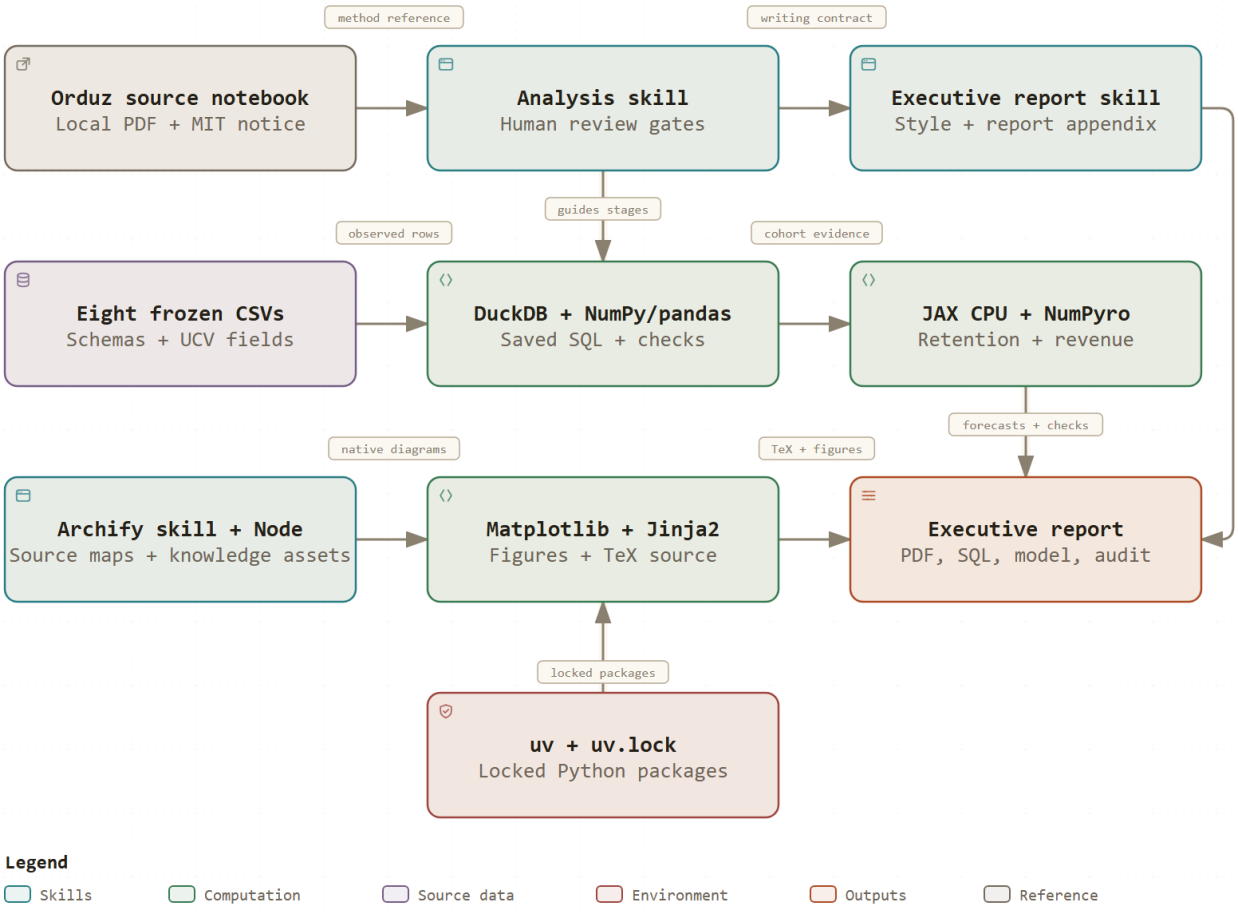


Figure 17. Analytical architecture. Knowledge assets and execution dependencies. Typed boxes, icons and colors distinguish skills, reference documents, Python packages, data and generated outputs.

THE SAME EVIDENCE SHOULD LEAD TO THE SAME REPORT

The recorded Python environment is installed with `uv sync --frozen`; commands run through `uv run --frozen`. The model uses CPU JAX/NumPyro; a GPU is not required. The analysis begins with the released records and does not regenerate them. Source hashes, saved SQL, model settings, figure data and document assets remain bound to the output manifests.

> Alternative models must use the same information available at the observation cutoff. Later outcomes are reserved for prospective evaluation and cannot inform features, parameter tuning or revisions to the input snapshot. Human review remains a gate between analytical stages.

Methods and source assets

Citations identify the source method and the scope of this implementation.

COHORT PRESENTATION

Bouhnick, G. (n.d.). *Cohort analysis—4 ways to analyze your product retention rate*. Mobilespoon. Original article (accessed October 29, 2026).

Inspiration for comparing cohort retention, revenue and acquisition recovery. All Studio North figures are computed from the released source records.

PERSISTENT CHURN HETEROGENEITY

Fader, P. S., and Hardie, B. G. S. (2007). How to project customer retention. *Journal of Interactive Marketing*, 21(1), 76–90. doi:10.1002/dir.20074.

Source of shifted-Beta-Geometric retention. The covariate regressions, cohort pooling and Bayesian priors are lab adaptations. An author-hosted working paper is also available.

RETENTION DYNAMICS AND CUSTOMER-BASE VALUE

Fader, P. S., and Hardie, B. G. S. (2010). Customer-base valuation in a contractual setting: The perils of ignoring heterogeneity. *Marketing Science*, 29(1), 85–93. doi:10.1287/mksc.1080.0482.

Motivates preserving survivor selection when valuing the remaining customer base. Article and technical appendix.

ACQUISITION, RETENTION AND SPENDING

McCarthy, D. M., Fader, P. S., and Hardie, B. G. S. (2017). Valuing subscription-based businesses using publicly disclosed customer data. *Journal of Marketing*, 81(1), 17–35. doi:10.1509/jm.15.0519.

Motivates the customer-base decomposition. This report estimates twelve-receipt contribution value and cash payback; it does not implement the paper's full corporate-valuation model.

Acquisition, forecasting and course connections

Sources distinguish descriptive cohort value, causal acquisition effects and the teaching objectives.

GAMMA REVENUE AND THE NEURAL COMPARISON

Orduz, J. C. (2024, January 8). *Cohort Revenue Retention Analysis with Flax and NumPyro*. Original article.

The main model adapts the Gamma revenue structure. The separate neural retention comparison adapts the architecture and weight priors to conditional renewal risk sets. Neither is a verbatim reproduction.

ATTRIBUTION, INCREMENTALITY AND ACQUISITION COST

Larsen, K. (2019, May 14). *You Don't Know Your Customer Acquisition Cost*. Noteworthy / The Journal Blog. Original article.

The assigned course reading motivates the counterfactual background inflow and average versus marginal acquisition costs. The experimental reconciliation and uncertainty calculations here are the lab's implementation. The article is not redistributed.

MBA 743 TEACHING CONNECTIONS

Tian, L. (2026). *MBA 743: Managing Customer Data*, Mod II. Kenan-Flagler Business School. Topic 1, Customer Journeys, slides 30–39; Topic 3, Acquisition Metrics, slides 6–10 and 23–34; Topic 4, Lifetime Value Metrics, slides 3, 30–31 and 35–42. Course slides and pre-lecture notes.

Topic 1 compares partitions of the same customers; Topic 3 links holdouts to incremental acquisition and contribution; Topic 4 distinguishes value before and after CAC and discounted payback. Course materials remain in the course distribution.

BUNDLED READING AND REPRODUCIBLE PROVENANCE

The GitHub Orduz asset folder contains the source-notebook PDF, the original notebook, its MIT notice and a hash manifest pinned to the source commit. The archived notebook is a reading asset; it was not rerun to generate the Studio North results. The repository's source guide records additional instructional background.